

PROBABILITY

**Course: Foundations of Statistics for Data Analytics and
Machine Learning Using Excel**

@ DV Data Analytics, Bangalore

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Counting

Counting

- Counting is a fundamental concept in combinatorics.
- Helps determine possible outcomes, optimize decisions, and solve real-world problems.
- Two key methods: **Permutations** and **Combinations**.
- Used in probability, statistics, Engineering and computer science.

The Counting Principle

- It states that if there are m ways to perform one action and n ways to perform another, then there are $m \times n$ ways to perform **both** actions.
- This principle is used to calculate the total number of possibilities when multiple events or choices are involved.
- Extends to multiple events: $m_1 \times m_2 \times \cdots \times m_k$.

Problem: A restaurant offers 3 choices of starters and 4 choices of main courses. How many different meal combinations can be made?

- Each appetizer choice can be paired with any main course choice.
- Using the multiplication principle there are $12(3 \times 4)$ different meal combinations.

Example: Counting Principle (Multiplication Principle)

Problem: How many different passwords can be created if a password consists of 3 letters (A-Z) followed by 2 digits (0-9)?

- Each letter has 26 choices.
- Each digit has 10 choices.
- Since each choice is independent, we multiply:

$$\begin{aligned}26 \times 26 \times 26 \times 10 \times 10 &= 26^3 \times 10^2 \\&= 17,576 \times 100 = 1,757,600\end{aligned}$$

Answer: There are 1,757,600 possible passwords.

Rules for Counting in Sets

Addition Principle:

If event A can occur in m ways and event B (which is independent of A) can occur in n ways, then the number of ways **either** event can occur is:

$$m + n$$

Example: A student should select one project and if that student can choose a project from two different categories:

- 3 AI projects
- 2 Robotics projects

Since the student cannot choose from both categories at the same time, the total number of choices is:

$$3 + 2 = 5$$

Limitations of addition and Multiplication rule: Choosing a Committee with Restrictions

A school is forming a 2-member committee from a pool of 4 students:

A, B, C, D

- Using Multiplication rule, we will get $4 \times 3 = 12$ different committees.
- The following are the committees AB, AC, AD, BC, BD, CD.
- Multiplication Rule Fails.

Incorrect Counting Example

Three officers—a president, a treasurer, and a secretary—are to be chosen from four people: Ann, Bob, Cyd, and Dan.

- Ann cannot be president.
- Either Cyd or Dan must be secretary.

A mistaken approach:

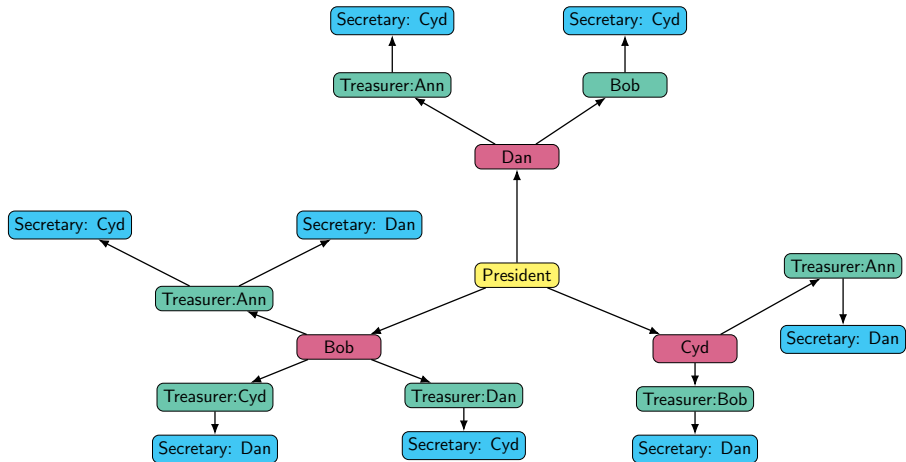
- 1 Choose a president from {Bob, Cyd, Dan} $\rightarrow$ 3 choices.
- 2 Choose a treasurer from the remaining 3 people $\rightarrow$ 3 choices.
- 3 Choose a secretary from {Cyd, Dan} $\rightarrow$ 2 choices.

Applying the multiplication principle incorrectly:

$$3 \times 3 \times 2 = 18$$

This answer is incorrect because choosing a treasurer should be from only 2 remaining candidates.

Officer Selection



Permutations

Permutations

Definition: The number of ways to arrange r objects from n distinct objects.

$${}^nP_r = P(n, r) = \frac{n!}{(n-r)!}$$

- Order matters.
- Example: Arranging 2 books from a set of 4.

All permutations of A,B,C,D:

- | | | |
|------|------|------|
| • AB | • BC | • CD |
| • AC | • BD | • DA |
| • AD | • CA | • DB |
| • BA | • CB | • DC |

$$P(4, 2) = \frac{4!}{2!} = \frac{24}{2} = \boxed{12}$$

Permutations with Repetition

Definition: The number of ways to arrange r objects from n objects when repetition is allowed.

$$P_{rep}(n, r) = n^r$$

- Order matters.
- Objects can be repeated.
- Example: Creating a 4-digit PIN using 10 digits (0-9):

$$P_{rep}(10, 4) = 10^4 = 10,000$$

Combinations

Combinations

Definition: The number of ways to choose r objects from n distinct objects without considering order.

$${}^nC_r = C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- Order does not matter.
- Example: Selecting 3 students from a group of 5.

All possible selections:

- | | | | |
|-------|-------|-------|-------|
| • ABC | • ACD | • BCD | • CDE |
| • ABD | • ACE | • BCE | |
| • ABE | • ADE | • BDE | |

$$C(5, 3) = \frac{5!}{3! \cdot 2!} = \frac{120}{6 \cdot 2} = \boxed{10}$$

Combinations with Repetition

Definition: The number of ways to choose r objects from n distinct objects when repetition is allowed.

$$C_{rep}(n, r) = \binom{n + r - 1}{r} = \frac{(n + r - 1)!}{r!(n - 1)!}$$

- Order does not matter.
- Objects can be repeated.
- Example: Choosing 3 scoops of ice cream from 5 flavors (where repeats are allowed):

$$C_{rep}(5, 3) = \binom{5 + 3 - 1}{3} = \binom{7}{3} = 35$$

Key Differences

- Permutations consider order; combinations do not.
- $P(n, r) > C(n, r)$ for $r > 1$.
- Use permutations when arranging, and combinations when selecting.

Example: Counting

Problem: A password consists of 4 distinct letters chosen from the English alphabet (26 letters). How many such passwords are possible?

Example: Counting

Problem: A password consists of 4 distinct letters chosen from the English alphabet (26 letters). How many such passwords are possible?

- Since order matters, we use permutations:

$$P(26, 4) = \frac{26!}{(26 - 4)!} = \frac{26!}{22!}$$

- Expanding the factorial:

$$26 \times 25 \times 24 \times 23 = 358800$$

- Thus, there are **358,800** possible passwords.

Summary Table with Excel Functions

	Formula	Without Repetition	With Repetition
Permutations	$P(n, r)$	$\frac{n!}{(n-r)!}$	n^r
Excel Formula		PERMUT(n, r)	PERMUTATIONA(n, r)
Combinations	$C(n, r)$	$\frac{n!}{r!(n-r)!}$	$\frac{(n+r-1)!}{r!(n-1)!}$
Excel Formula		COMBIN(n, r)	COMBINA(n, r)

Excel Functions for Counting and Probability

Function	Description	Example	Output
RAND()	Generates a random number between 0 and 1	=RAND()	0.423 (varies)
RANDBETWEEN	Generates a random integer between two bounds	=RANDBETWEEN(1, 6)	4 (varies)
FACT	Returns the factorial of a number	=FACT(5)	120
PERMUT	Returns the number of permutations (order matters)	=PERMUT(5, 3)	60
PERMUTATIONA	Returns the number of permutations allowing repetitions	=PERMUTATIONA(5, 3)	125
COMBIN	Returns the number of combinations (order does not matter)	=COMBIN(5, 3)	10
COMBINA	Returns the number of combinations allowing repetitions	=COMBINA(5, 3)	35
ROUND	Rounds a number to a specified number of digits	=ROUND(0.123456, 2)	0.12

Probability

Introduction to Probability

- **Probability** of an event is a measure (number) of the chance with which we can expect the event to occur.
- It is quantified as a number between 0 and 1.
- Defined as:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

- Probability values range from 0 (impossible event) to 1 (certain event).

Introduction to Probability

Basic Concepts:

- **Outcome:** A single possible result of a random experiment. It is an element of the sample space S .
- **Sample Space (S):** The set of all possible outcomes.
- **Event (E):** A subset of the sample space.
- **Probability of an event:** $P(E) = \frac{|E|}{|S|}$ (for equally likely outcomes).

Example: Rolling a Die

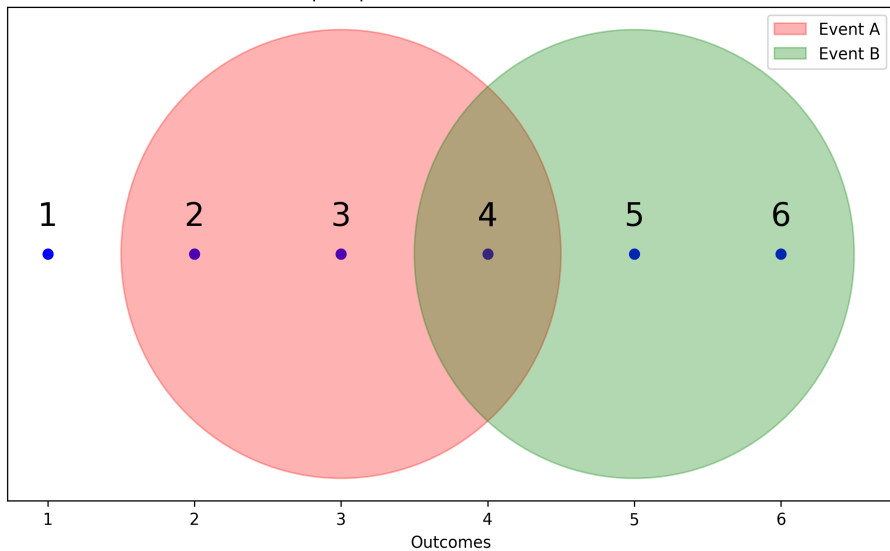
- Sample space: $S = \{1, 2, 3, 4, 5, 6\}$.
- Outcome: Rolling a 3 (i.e., outcome = 3).
- Event A : Rolling an even number, $A = \{2, 4, 6\}$.

Key Difference:

- An outcome is a single result.
- An event consists of multiple possible outcomes.

Outcome, Sample Space and Event

Sample Space, Events, and Outcomes



Classical Probability (Theoretical Probability)

- Based on equally likely outcomes.
- Formula: $P(E) = \frac{M}{N}$, where M is the number of favorable outcomes and N is the total outcomes.
- **Example:** Rolling a fair six-sided die. The probability of rolling a 4:

$$P(4) = \frac{1}{6}$$

Empirical Probability (Experimental Probability)

- Based on observed data from experiments.
- Formula: $P(E) = \frac{\text{Number of times } E \text{ occurs}}{\text{Total number of trials}}$.
- **Example:** If a coin is flipped 100 times and heads appear 55 times:

$$P(H) = \frac{55}{100} = 0.55$$

Subjective Probability

- Based on intuition, personal judgment, or expert opinions.
- Used when no historical data is available.
- **Example:** A doctor estimates a 70% chance of recovery for a patient based on symptoms and experience.

When Does an Event Occur?

Question: If one outcome in an event happens, does the event occur?

Answer: Yes! If at least one outcome from the event occurs, then the event has occurred.

Example: Rolling a Die

- Event A : Rolling an even number, $A = \{2, 4, 6\}$.
- If the outcome is 4, then since $4 \in A$, event A occurs.
- If the outcome is 3, then since $3 \notin A$, event A does not occur.

Conclusion: An event occurs if any of its outcomes happens.

Why Permutations and Combinations in Probability?

- Probability is about comparing:

$$\text{Probability} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

- To find these counts, we often need:
 - **Permutations** – when *order matters*
 - **Combinations** – when *order doesn't matter*

Example 1: Permutation (Order Matters)

Problem: A password consists of 3 different digits from 0–9. What is the probability that you guess it correctly on the first try?

Solution:

- Total outcomes = Number of ways to arrange 3 digits from 10:

$${}^{10}P_3 = \frac{10!}{(10-3)!} = 720$$

- Favorable outcomes = 1 (only one correct password)

$$\text{Probability} = \frac{1}{720}$$

Example 2: Combination (Order Doesn't Matter)

Problem: From a group of 8 students, a committee of 3 is selected. What is the probability that a specific group of 3 students is chosen?

Solution:

- Total combinations:

$$\binom{8}{3} = \frac{8!}{3! \cdot 5!} = 56$$

- Favorable outcomes = 1 (only one specific group)

$$\text{Probability} = \frac{1}{56}$$

Axioms and Rules of Probability

Axioms of Probability

Given a sample space S and an event E , probability satisfies:

- 1 Non-negativity: $P(E) \geq 0$ for all events E .
- 2 Normalization: $P(S) = 1$ (Sum of probabilities of all events = 1).
- 3 Additivity: If $E_1, E_2, \dots$ are disjoint events, then
$$P(E_1 \cup E_2 \cup \dots) = P(E_1) + P(E_2) + \dots$$

Example: Rolling a fair die

- $S = \{1, 2, 3, 4, 5, 6\}$
- $P(\text{rolling a 2}) = \frac{1}{6}$
- $P(\text{rolling an even number}) = P(2) + P(4) + P(6) = \frac{3}{6} = \frac{1}{2}$

Basic Probability Rules

- **Addition Rule:** If events A and B are mutually exclusive:

$$P(A \cup B) = P(A) + P(B)$$

- **Multiplication Rule:** If events A and B are independent:

$$P(A \cap B) = P(A) \times P(B)$$

- **Complement Rule:** The probability of an event not occurring:

$$P(A^c) = 1 - P(A)$$

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$$P(2 \text{ or } 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

- *Example:* Tossing a coin and rolling a die. Probability of getting heads **and** a 4:

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$$P(H \text{ and } 4) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

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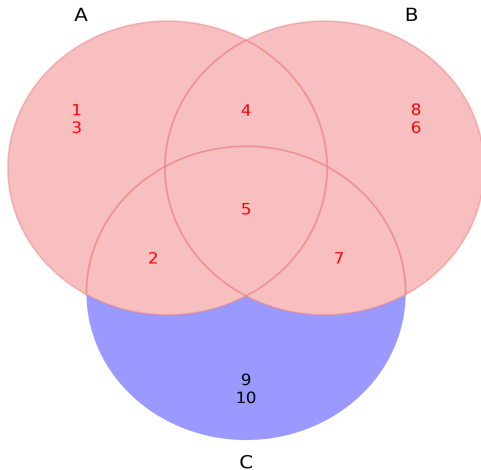
$$P(\text{not } 6) = 1 - \frac{1}{6} = \frac{5}{6}$$

Union of sets: Addition Rule

A: {1, 2, 3, 4, 5}

B: {4, 5, 6, 7, 8}

Venn Diagram Showing $A \cup B$ with Three Sets



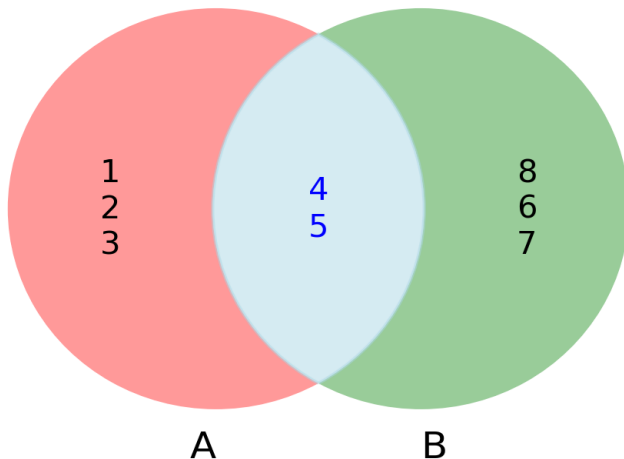
C: {2, 5, 7, 9, 10}

Intersection of sets: Multiplication Rule

Venn Diagram Showing $A \cap B$

A: {1, 2, 3, 4, 5}

B: {4, 5, 6, 7, 8}

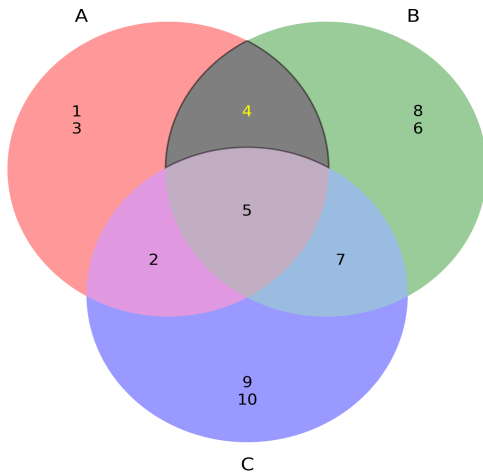


Intersection of sets

A: {1, 2, 3, 4, 5}

B: {4, 5, 6, 7, 8}

Venn Diagram Showing Intersection of A and B

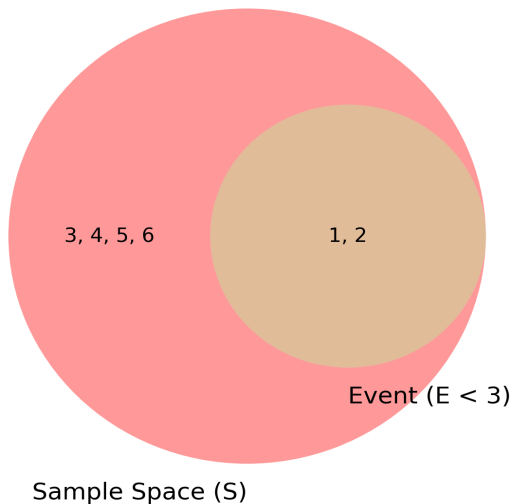


C: {2, 5, 7, 9, 10}

Sample Space and Event

- Event is a subset of Sample Space.

Venn Diagram of Sample Space and Event ($E < 3$)



Mutually Exclusive Events

Mutually Exclusive Events

Definition: Two events A and B are said to be **mutually exclusive** if they cannot occur at the same time.

Key Characteristics:

- If one event happens, the other cannot.
- Their intersection is empty: $A \cap B = \emptyset$.
- The probability of both occurring together is zero.

$$P(A \cap B) = 0.$$

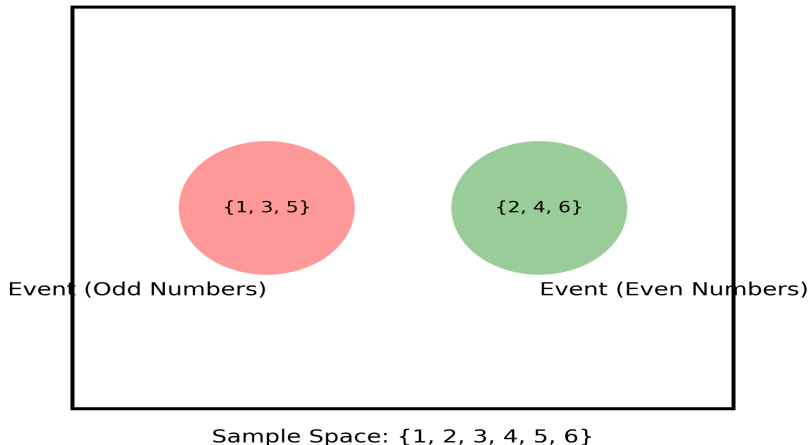
Example: Rolling a Die

- Let A be the event of rolling an even number: $A = \{2, 4, 6\}$.
- Let B be the event of rolling an odd number: $B = \{1, 3, 5\}$.
- Since no number can be both even and odd, A and B are mutually exclusive.
- Mathematically, $P(A \cap B) = 0$.

Sample Space and Mutually Exclusive Events

- Events that cant occur at the same time.

Venn Diagram of Sample Space and Mutually Exclusive Events



Conditional Probability

Definition: The conditional probability of an event A given that event B has occurred is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Key Points:

- Measures the likelihood of A occurring given that B has occurred.
- The probability space is reduced to B .
- Requires that $P(B) > 0$.

Example Problem

Example: A box contains 5 red balls and 7 blue balls. If a ball is drawn at random, what is the probability that it is red given that it is not blue?

Solution:

- Let A be the event that the ball drawn is red.
- Let B be the event that the ball drawn is not blue.
- Since the only non-blue balls are red, we have $P(A \cap B) = P(A)$.

Using the conditional probability formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{5/12}{5/12} = 1.$$

Independence

Independent Events

Definition: Two events A and B are said to be **independent** if the occurrence of one does not affect the probability of the other.

Mathematically,

$$P(A \cap B) = P(A)P(B).$$

Equivalent Conditions:

- $P(A | B) = P(A)$, meaning knowing that B occurred does not change the probability of A .
- $P(B | A) = P(B)$, meaning knowing that A occurred does not change the probability of B .

Example: Suppose we roll a fair die and flip a fair coin. Define:

- A : Getting a 6 on the die.
- B : Getting heads on the coin.

Since rolling the die does not affect the coin flip,

$$P(A \cap B) = \left(\frac{1}{6}\right) \times \left(\frac{1}{2}\right) = \frac{1}{12},$$

which matches $P(A)P(B)$, proving independence. □

Examples and Misconceptions

Example: Drawing Cards from a Deck

- Let A be the event of drawing a red card.
- Let B be the event of drawing a spade.
- Since a card cannot be both red and a spade, $P(A \cap B) = 0$.
- However, $P(A)P(B) = \left(\frac{26}{52} \times \frac{13}{52}\right) \neq 0$.
- Since $P(A \cap B) \neq P(A)P(B)$, A and B are **not independent**.

Common Misconceptions:

- **Independence is not the same as mutual exclusivity:** If A and B are mutually exclusive, then $P(A \cap B) = 0$, so they cannot be independent unless $P(A) = 0$ or $P(B) = 0$.
- **Independence is not about equal probabilities:** Events can be independent even if they have different probabilities.

Understanding $P(A \cap B)$: Intersection of Events

Case	Formula and Explanation
Independent Events	$P(A \cap B) = P(A) \cdot P(B)$ A and B do not affect each other (e.g., coin toss and die roll)
Dependent Events	$P(A \cap B) = P(A) \cdot P(B A)$ Probability of B depends on A happening (e.g., drawing 2 cards without replacement)
Mutually Exclusive Events	$P(A \cap B) = 0$ A and B cannot happen at the same time (e.g., getting 2 and 5 on a single die roll)

Law of Total Probability

Law of Total Probability

- The **law of total probability** is a fundamental rule in probability theory.
- The law of total probability provides a way to compute the probability of an event by considering all possible ways that event can occur.
- It provides a way to compute the probability of an event by considering partitioning of the sample space.
- Useful when dealing with conditional probabilities.
- If $B_1, B_2, \dots, B_n$ form a partition of the sample space and $P(B_i) > 0$ for all i , then for any event A :

$$P(A) = \sum_{i=1}^n P(A \mid B_i)P(B_i)$$

Illustration

Consider a sample space partitioned into disjoint events $B_1, B_2, B_3 \dots B_8$:

- Each B_i represents a different way an event can happen.
- The probability of A is the sum of its conditional probabilities weighted by the probability of each B_i .



Example 1: Defective Items

A factory has two machines:

- Machine 1 produces 60% of items, and Machine 2 produces 40%.
- Machine 1 has a 5% defect rate, while Machine 2 has a 10% defect rate.

What is the probability that a randomly chosen item is defective?

Solution

Define events:

- D : Item is defective
- B_1 : Item from Machine 1, $P(B_1) = 0.6$
- B_2 : Item from Machine 2, $P(B_2) = 0.4$

Given:

$$P(D | B_1) = 0.05, \quad P(D | B_2) = 0.10$$

Using the law of total probability:

$$P(D) = P(D | B_1)P(B_1) + P(D | B_2)P(B_2)$$

$$P(D) = (0.05)(0.6) + (0.10)(0.4) = 0.03 + 0.04 = 0.07$$

Thus, the probability of selecting a defective item is 0.07 (or 7%).

Bayes Rule

Bayes Theorem

- **Bayes Theorem** is a fundamental result in probability theory.
- It describes how to update our belief about an event based on new evidence.
- Commonly used in medical diagnosis, spam filtering, and machine learning.

Derivation of Bayes' Theorem:

- Consider a sample space partitioned into two mutually exclusive events B_1 and B_2 , such that $B_1 \cup B_2 = S$.
- From the definition of conditional probability:

$$P(A | B_1) = \frac{P(A \cap B_1)}{P(B_1)}$$

$$P(B_1 | A) = \frac{P(A \cap B_1)}{P(A)}$$

- Rearranging both equations:

$$P(A \cap B_1) = P(A | B_1)P(B_1)$$

$$P(A \cap B_1) = P(B_1 | A)P(A)$$

- Equating the two expressions for $P(A \cap B_1)$:

$$P(A | B_1)P(B_1) = P(B_1 | A)P(A)$$

- Solving for $P(B_1 | A)$ gives:

$$P(B_1 | A) = \frac{P(A | B_1)P(B_1)}{P(A)}$$

- This is Bayes' Theorem.
- Using the Law of Total Probability:

$$P(A) = P(A | B_1)P(B_1) + P(A | B_2)P(B_2)$$

Theorem Statement

Bayes' Theorem: Given an event A and a partition of the sample space $B_1, B_2, \dots, B_n$:

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{P(A)}$$

where

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$$

LTP: If $B_1, B_2, \dots, B_n$ form a partition of the sample space, then for any event A :

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$$

- This rule helps compute the probability of an event by considering all possible ways it can occur.
- It is used as the denominator in Bayes' Theorem.

Example 1: Spam Filtering

- A spam filter detects spam emails based on keywords.
- Suppose 20% of emails are spam.
- Given
- Given that an email contains the word "free", how likely is it to be spam?

Let:

$$\begin{aligned}P(S) &= 0.20, & P(S^c) &= 0.80 \\P(F | S) &= 0.50, & P(F | S^c) &= 0.05\end{aligned}$$

Using Bayes' Theorem:

$$P(S | F) = \frac{P(F | S)P(S)}{P(F)}$$

$$P(F) = (0.50 \times 0.20) + (0.05 \times 0.80) = 0.14$$

$$P(S | F) = \frac{0.50 \times 0.20}{0.14} = 0.714$$

The probability that an email is spam given that it contains "free" is

Bayes' Theorem and its Relationship with LTP

- **LTP** computes the total probability of an event.
- **Bayes' Theorem** inverts this relationship to find the probability of causes given evidence.
- It allows us to update beliefs based on new information.

Law of Total Probability vs Bayes' Rule

Big Picture: How They Differ

Pizza Delivery Example

- 60% orders from A (20% late), 40% from B (50% late)

Law of Total Probability:

$$P(\text{Late}) = P(\text{Late} \mid A) \cdot P(A) + P(\text{Late} \mid B) \cdot P(B)$$

→ *What's the chance your pizza is late overall?*

Bayes' Rule:

$$P(B \mid \text{Late}) = \frac{P(\text{Late} \mid B) \cdot P(B)}{P(\text{Late})}$$

→ *Given that it's late, what's the chance it came from B?*

Law of Total Probability

Think of it as...	Looking forward from causes to outcomes
What it helps with	Calculates overall probability of an outcome from multiple scenarios

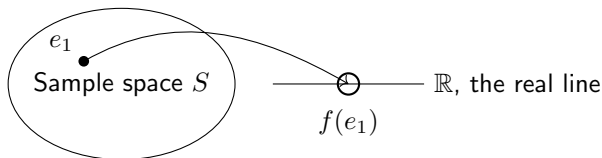
Bayes' Rule

Think of it as...	Looking backward from outcomes to causes
What it helps with	Updates the probability of a cause given that an outcome has occurred

Random Variables

Introduction to Random Variables

- A **random variable** is a numerical description of the outcome of a statistical experiment. It is a way to assign numbers to outcomes of a random experiment
- It is a function defined on a sample space, S , that associates a real number with each outcome in S .



- Two main types:
 - **Discrete Random Variables:** Take countable values (e.g., number of heads in a coin toss).
 - **Continuous Random Variables:** Take an infinite number of values within a range (e.g., height of individuals).
- If each event is associated with its probability in random variable, it is called as **Probability distribution**.

Example: Coin Toss

- Experiment: Toss a coin
- Let X be a random variable such that:
 - $X = 1$ if Heads
 - $X = 0$ if Tails
- X is a discrete random variable.

What is a Probability Distribution?

- A **probability distribution** tells us the **probability** of each value of a random variable.
- Shows how likely each outcome is.
- A distribution describes how values of a random variable are spread.
- It helps in understanding the likelihood of different outcomes.
- Examples: Normal, Uniform, Binomial, Poisson, etc.
- Can be a:
 - **Probability distribution** (for discrete variables)
 - **Probability density function** (for continuous variables)

Creating a Distribution:

- Consider rolling a six-sided die multiple times.
- We record the outcomes and plot their frequency.

Coin Toss: Distribution Example

Outcome	Value (X)	Probability
Heads	1	0.5
Tails	0	0.5

- This is the probability distribution of X .
- X is a **Random Variable**

Dice Roll Example

- Experiment: Roll a fair 6-sided die
- Random variable X = number on the die

Value of X	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

This is a uniform distribution.

Summary

Concept	Explanation
Random Variable	A variable that assigns numbers to outcomes of a random process
Distribution	Describes how probabilities are assigned to each value of the random variable

- **Probability Distribution** assigns **probabilitiy** to **each outcome** of a **random variable**.
- A probability distribution links every value (or interval) of a random variable to its probability of occurring, forming the basis for all statistical inference.

Example

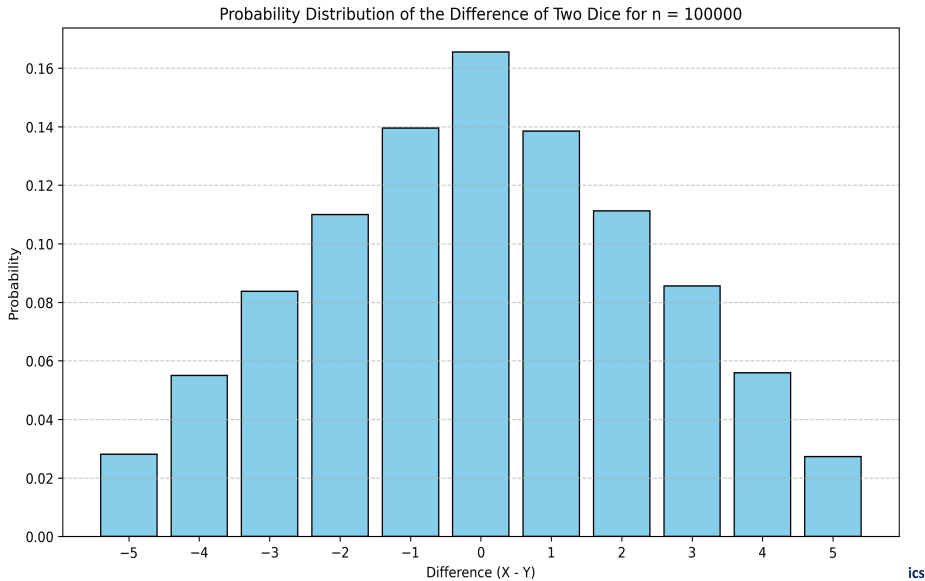
Let X and Y be the outcomes of two fair 6-sided dice.
Define a new **random variable**:

$$D = X - Y$$

Possible values of the **random variable** D are:

$$\{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$$

The Distribution



Tabulated Distribution (Example)

D	Probability	Cumulative
-5	0.0278	0.0278
-4	0.0556	0.0834
-3	0.0833	0.1667
-2	0.1111	0.2778
-1	0.1389	0.4167
0	0.1667	0.5834
1	0.1389	0.7223
2	0.1111	0.8334
3	0.0833	0.9167
4	0.0556	0.9723
5	0.0278	1.0000

- Observe the Cumulative Probability

Motivating a Formal Function

Question: How do we represent a distribution mathematically?

Answer: Using a function that tells us the probability of each value:

- For discrete random variables: **Probability Mass Function (PMF)**
- For continuous random variables: **Probability Density Function (PDF)**

PMF: Probability Mass Function

Definition: For a discrete random variable X , the PMF is defined as:

$$p(x) = P(X = x)$$

Example: Let $D = X - Y$ be the difference of two dice rolls. Then:

$$p(0) = 0.1667, \quad p(1) = 0.1389, \quad \dots$$

Properties:

- $0 \leq p(x) \leq 1$
- $\sum_x p(x) = 1$

PDF: Probability Density Function

Definition: For a continuous random variable X , the PDF is a function $f(x)$ such that:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties:

- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x) dx = 1$
- It means The total area under the PDF curve is always 1

Note: $f(x)$ is not a probability — only areas under the curve give probabilities.

PMF vs PDF

- **PMF:** Used for discrete variables (e.g., dice outcomes)
- **PDF:** Used for continuous variables (e.g., height, weight)
- Both describe how probability is distributed over outcomes

Aspect	Discrete Distribution	Continuous Distribution
Type of Values	Countable outcomes (e.g., 0, 1, 2)	Uncountable outcomes (e.g., real numbers)
Key Function	Probability Mass Function (PMF)	Probability Density Function (PDF)
Formula	$P(X = x_i) = p(x_i)$	$P(a \leq X \leq b) = \int_a^b f(x) dx$
Total Probability	$\sum_i p(x_i) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
Probability at Exact Value	Non-zero: $P(X = x) > 0$	Always zero: $P(X = x) = 0$

Why CDF?

Question: What is the probability that a variable is *less than or equal to* a value?

Answer: We use the **Cumulative Distribution Function (CDF)** to capture that.

Cumulative Distribution Function (CDF)

Definition: For a random variable X , the CDF is defined as:

$$F(x) = P(X \leq x)$$

- **Discrete:** $F(x) = \sum_{t \leq x} p(t)$
- **Continuous:** $F(x) = \int_{-\infty}^x f(t) dt$

Properties of CDF

- $F(x)$ is non-decreasing
- $\lim_{x \rightarrow -\infty} F(x) = 0$
- $\lim_{x \rightarrow \infty} F(x) = 1$
- For continuous X , $F'(x) = f(x)$

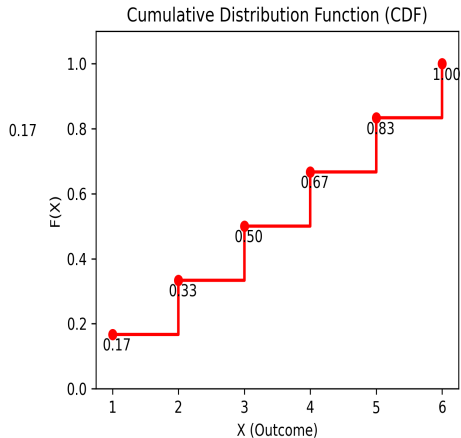
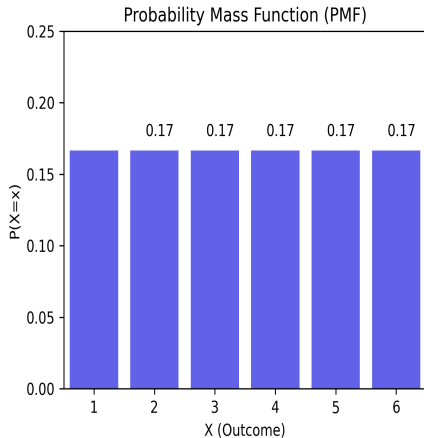
Example: CDF of $D = X - Y$

- We already saw the PMF for D .
- The CDF at a point is the sum of all probabilities up to that point.

Example:

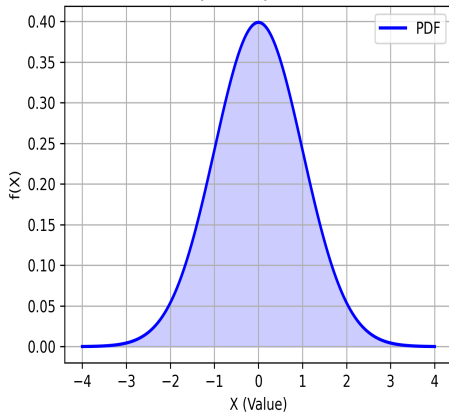
$$F(0) = P(D \leq 0) = p(-5) + p(-4) + \cdots + p(0) = 0.5834$$

PMF and CDF

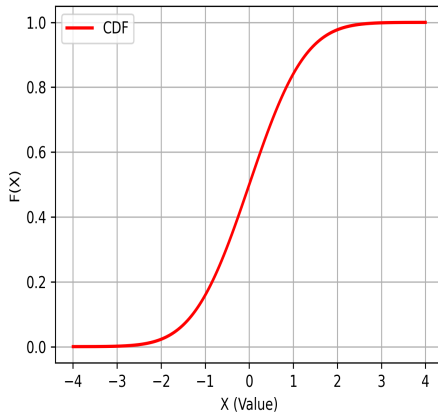


PDF and CDF

Probability Density Function (PDF)



Cumulative Distribution Function (CDF)



Comparison of PMF, PDF, and CDF

- **Probability Mass Function (PMF):**

- Used for discrete random variables.
- Assigns probabilities to specific values.

- **Probability Density Function (PDF):**

- Used for continuous random variables.
- Represents density, not direct probabilities.
- The area under the curve represents probability.

- **Cumulative Distribution Function (CDF):**

- Used for both discrete and continuous random variables.
- Gives cumulative probability up to a given value.
- Always non-decreasing and ranges from 0 to 1.

Why Do We Need the CDF When We Have the PDF?

- A **PDF** (Probability Density Function) shows how likely values are **around a point**.
- But in continuous data, the chance of getting **one exact value is always 0**.
- We want to know: *"What is the probability that $X \leq x$?"*
- The **CDF** (Cumulative Distribution Function) gives us that total probability.

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

PDF vs CDF – What They Tell Us

Question	PDF	CDF
What is the shape of the data?	Yes	No
Chance up to a value ($P(X \leq x)$)?	No	Yes
Probability in a range ($P(a \leq X \leq b)$)?	Needs integration	$F(b) - F(a)$
Exact value ($P(X = x)$)?	Always 0	Always 0

Example: Rolling a Die

- Consider rolling a fair six-sided die.
- The outcome X is a discrete random variable with possible values: $\{1, 2, 3, 4, 5, 6\}$.
- Probability Mass Function (PMF):

$$P(X = x) = \frac{1}{6}, \quad x = 1, 2, 3, 4, 5, 6$$

- Expected Value:

$$E[X] = \sum xP(X = x) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

Joint Distribution: Definition

Definition

The **joint distribution** of two random variables X and Y gives the probability that X takes a specific value and Y takes a specific value at the same time.

Notation: $P(X = x, Y = y)$

Example: Joint Distribution Table

$X \backslash Y$	1	2	3
1	0.1	0.1	0.1
2	0.2	0.1	0.1
3	0.05	0.1	0.15

Each cell gives $P(X = x, Y = y)$.

Marginal Distribution: Definition

Definition

The **marginal distribution** of a random variable is the probability distribution of that variable alone, obtained by summing (or integrating) the joint distribution over the other variable.

For discrete variables:

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$P(Y = y) = \sum_x P(X = x, Y = y)$$

Finding Marginal Distributions (Example)

From the previous table:

Marginal of X :

$$P(X = 1) = 0.1 + 0.1 + 0.1 = 0.3$$

$$P(X = 2) = 0.2 + 0.1 + 0.1 = 0.4$$

$$P(X = 3) = 0.05 + 0.1 + 0.15 = 0.3$$

Marginal of Y is found similarly by adding down columns.

Joint and Marginal Densities (Continuous Case)

Joint Probability Density Function (pdf): $f(x, y)$

- $f(x, y)$ satisfies:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

- Probability that (X, Y) lies in a region A :

$$P((X, Y) \in A) = \iint_A f(x, y) dx dy$$

Marginal Densities:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Key Points

- **Joint Distribution** shows relationship between two random variables.
- **Marginal Distribution** looks at one variable alone.
- In the discrete case, sum probabilities; in the continuous case, integrate.
- Helps in finding conditional distributions and checking independence.

Quick Check!

Question:

Given the joint table below:

$X \backslash Y$	0	1
0	0.3	0.2
1	0.1	0.4

Find:

- $P(X = 0)$
- $P(Y = 1)$

Marginal for $X = 0$:

$$P(X = 0) = 0.3 + 0.2 = 0.5$$

Marginal for $Y = 1$:

$$P(Y = 1) = 0.2 + 0.4 = 0.6$$

Independence of Random Variables

Definition

Two random variables X and Y are **independent** if and only if:

$$P(X = x, Y = y) = P(X = x) \times P(Y = y) \quad \text{for all } x, y$$

Otherwise, X and Y are said to be **dependent**.

Idea Behind Checking Independence

- Start from the **joint probability** $P(X = x, Y = y)$.
- Compute the **marginal probabilities** $P(X = x)$ and $P(Y = y)$.
- Check if:

$$P(X = x, Y = y) = P(X = x) \times P(Y = y)$$

for all possible pairs (x, y) .

Example: Joint Distribution Table

X\Y	0	1
0	0.3	0.2
1	0.1	0.4

This table gives $P(X = x, Y = y)$ for each pair (x, y) .

Step 1: Marginal Probabilities

Compute Marginal of X :

$$P(X = 0) = 0.3 + 0.2 = 0.5$$

$$P(X = 1) = 0.1 + 0.4 = 0.5$$

Compute Marginal of Y :

$$P(Y = 0) = 0.3 + 0.1 = 0.4$$

$$P(Y = 1) = 0.2 + 0.4 = 0.6$$

Step 2: Check for Independence

Check if $P(X = x, Y = y) = P(X = x) \times P(Y = y)$ for all x, y .

Example:

$$P(X = 0, Y = 0) = 0.3$$

$$P(X = 0) \times P(Y = 0) = 0.5 \times 0.4 = 0.2$$

Since $0.3 \neq 0.2$, X and Y are NOT independent.

Conclusion

- If $P(X = x, Y = y) \neq P(X = x)P(Y = y)$ for any (x, y) , variables are **dependent**.
- Independence implies that knowing X tells us nothing about Y , and vice versa.

Checking joint vs marginal probabilities is the key!

Independence and Conditional Distributions

Independence of Random Variables

Two random variables X and Y are **independent** if:

$$P(X = x, Y = y) = P(X = x) \times P(Y = y) \quad \text{for all } x, y$$

Interpretation: Knowing X does not affect beliefs about Y .

Effect on Conditional Distributions

If independent:

$$P(Y = y \mid X = x) = P(Y = y)$$

(Conditioning on X does not change the distribution of Y .)

If not independent:

$$P(Y = y \mid X = x) \neq P(Y = y)$$

(Knowing X *does* change the distribution of Y .)

Independence vs Dependence: A Comparison

Property	Independent	Dependent
Joint Probability	$P(X, Y) = P(X)P(Y)$	$P(X, Y) \neq P(X)P(Y)$
Conditional Probability	$P(Y X) = P(Y)$	$P(Y X)$ changes with X
Interpretation	X and Y do not affect each other	X and Y influence each other

Key Point:

Independence means the conditional distribution is the same as the marginal distribution.

Probability Distributions: Binomial Distributions

Binomial Distribution

- A **Bernoulli trial** is an experiment or process that results in a binary outcome: success (usually coded as 1) or failure (usually coded as 0).
- The **binomial distribution** describes the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success.
- The binomial distribution can be defined by the following parameters:
 - n : The number of trials.
 - p : The probability of success on a single trial.
 - k : The number of successes in n trials.
- The probability mass function (PMF) of the binomial distribution is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient.
- p^k is the probability of having k successes.
- $(1 - p)^{n-k}$ is the probability of having $n - k$ failures.

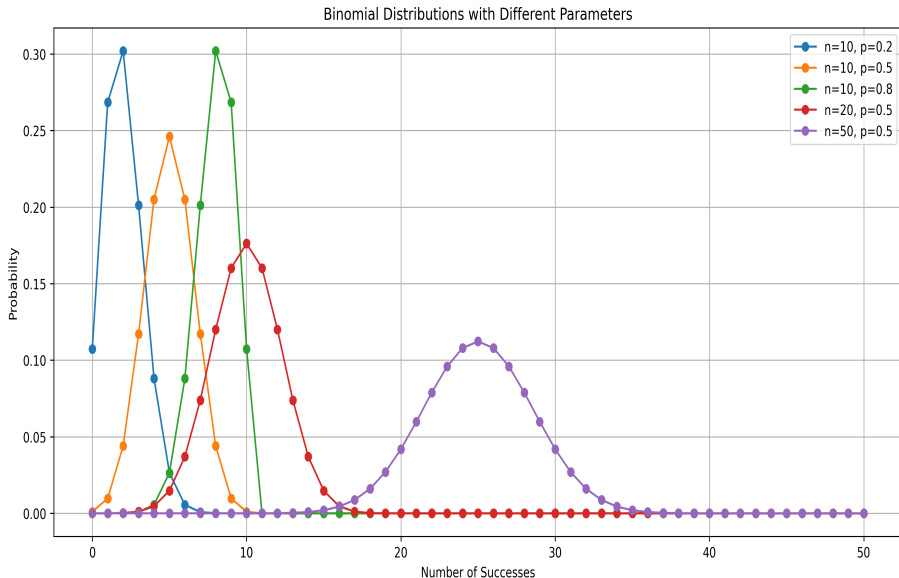
Properties of the Binomial Distribution

Some important properties of the binomial distribution:

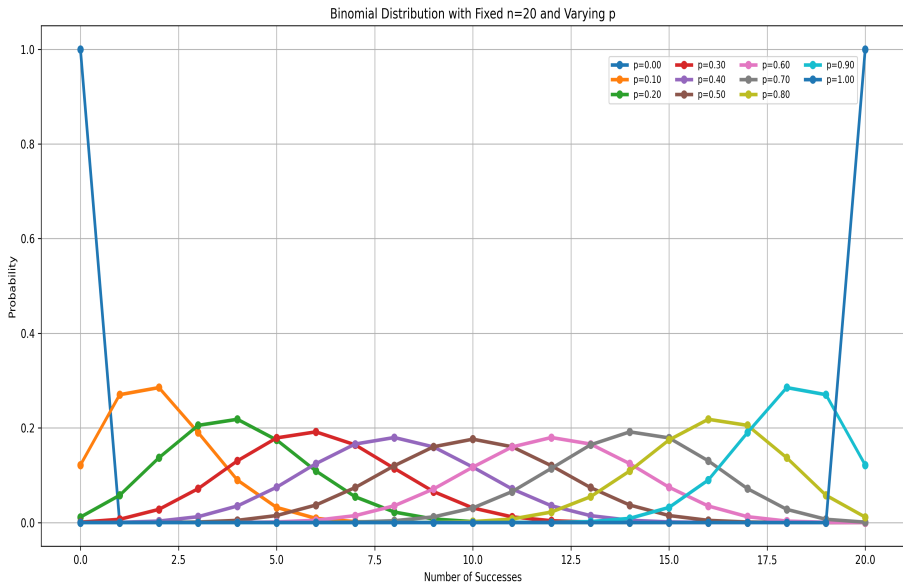
- **Mean:** $\mu = n \cdot p$
- **Variance:** $\sigma^2 = n \cdot p \cdot (1 - p)$
- **Standard Deviation:** $\sigma = \sqrt{n \cdot p \cdot (1 - p)}$

These properties help in understanding the central tendency and spread of the distribution.

Binomial Distribution for different parameters



Binomial Distribution for different parameters



Example Problem

Consider a scenario where a fair coin is flipped 10 times. What is the probability of getting exactly 6 heads?

We know the following:

- $n = 10$ (number of flips),
- $p = 0.5$ (probability of getting a head),
- $k = 6$ (number of heads we are interested in).

The probability mass function for the binomial distribution is:

$$P(X = 6) = \binom{10}{6} (0.5)^6 (0.5)^{10-6} = \binom{10}{6} (0.5)^{10}$$

Now, calculating the binomial coefficient:

$$\binom{10}{6} = \frac{10!}{6!4!} = 210$$

Therefore, the probability is:

$$P(X = 6) = 210 \times (0.5)^{10} = 210 \times \frac{1}{1024} \approx 0.205$$

Probability Distributions: Poisson Distributions

What is the Poisson Distribution?

- This distribution gives the probability of a given number of events occurring in a fixed interval of time or space.
- It needs the average number of times the event occurs over that interval (λ).
- It is typically used for events that happen independently of each other.

Characteristics:

- The events are independent.
- The rate (λ) of occurrence is constant.
- The number of events in non-overlapping intervals are independent.

The Poisson distribution is often used in situations like:

- Number of phone calls at a call center in an hour.
- Number of accidents at a traffic intersection in a day.
- Number of customers arriving at a store in a given time period.

Poisson Distribution Formula

The probability mass function (PMF) of the Poisson distribution is given by:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

where:

- X is the random variable representing the number of events.
- λ is the average number of events in the given time period (also called the rate parameter).
- k is the number of events observed.
- e is the Euler's number, approximately 2.71828.

This formula gives the probability of observing exactly k events when the expected number of events is λ .

Mean and Variance of Poisson Distribution

For a Poisson distribution with parameter λ :

- The mean is:

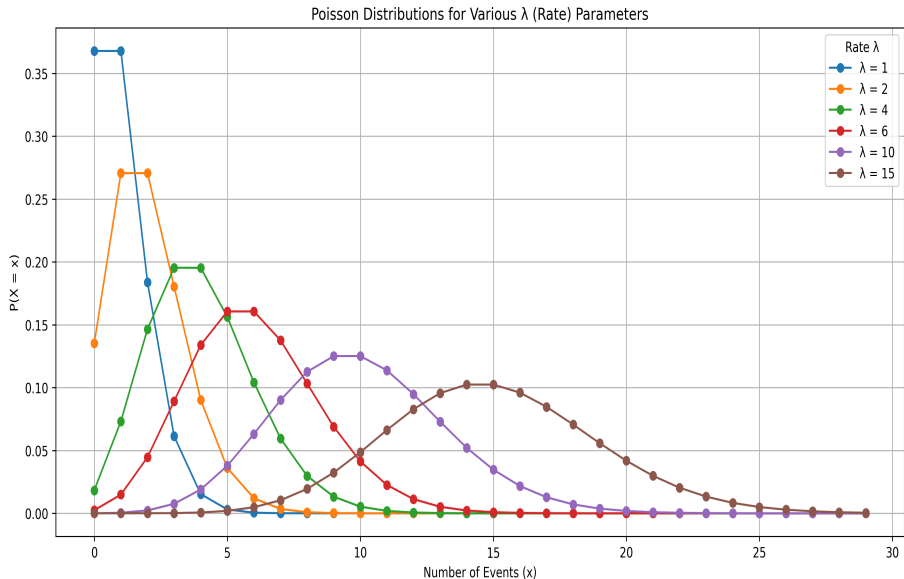
$$\mu = \lambda$$

- The variance is:

$$\sigma^2 = \lambda$$

Notice that the mean and variance of a Poisson distribution are both equal to λ , which is an interesting property.

Poisson Distribution for different parameters



Example of Poisson Distribution

- The **average number of cars** passing through a toll booth **per hour** is 5
- **Calculate** the probability of exactly 3 cars passing through the booth in one hour.
- Here, the rate $\lambda = 5$ (the average number of cars per hour).
- **Find** the probability when $k = 3$.

Using the Poisson formula:

$$P(X = 3) = \frac{5^3 e^{-5}}{3!}$$

$$P(X = 3) = \frac{125e^{-5}}{6}$$

$$P(X = 3) \approx 0.1404$$

So, the probability of exactly 3 cars passing through the toll booth in an hour is approximately 0.1404, or 14.04%.

Applications of Poisson Distribution

The Poisson distribution is widely used in various fields, such as:

- **Telecommunications:** Number of calls or messages in a given time frame.
- **Healthcare:** Number of patients arriving at an emergency room in a given hour.
- **Traffic Flow:** Number of cars passing a point in a road during a fixed period.
- **Physics:** Number of radioactive decay events in a time period.

The Poisson distribution is particularly useful when events occur randomly and independently, and we are interested in counting how often they happen in a specific interval.

Probability Distributions: Normal Distributions

What is the Normal Distribution?

- The Normal distribution is a continuous probability distribution that is symmetric about the mean.
- Its natural occurrence is in many real-world phenomena, hence called as **normal**.

Key Characteristics:

- It has a bell-shaped curve.
- The distribution is symmetric around the mean (μ).
- The mean, median, and mode of the distribution are all the same.
- The standard deviation (σ) controls the spread of the distribution.
- It is defined by two parameters: the mean (μ) and the standard deviation (σ).

The Normal distribution is used to model many natural and social phenomena such as heights, test scores, and errors in measurements.

Normal Distribution Formula

The probability density function (PDF) of the Normal distribution is given by:

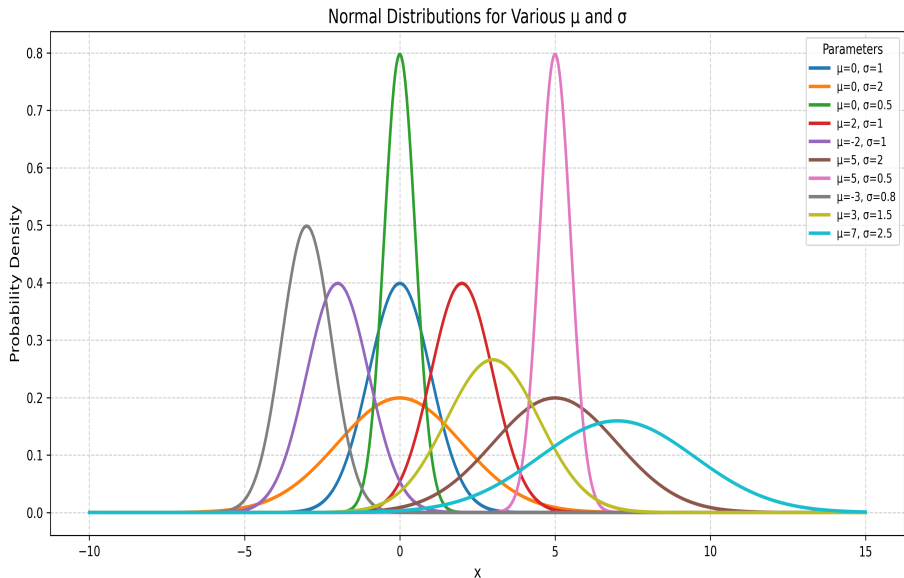
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- $f(x)$ is the probability density at x .
- μ is the mean (average) of the distribution.
- σ is the standard deviation (a measure of the spread of the distribution).
- x is the variable of interest.
- e is Euler's number (approximately 2.71828).

The Normal distribution is fully described by its mean and standard deviation.

Normal Distribution for different parameters



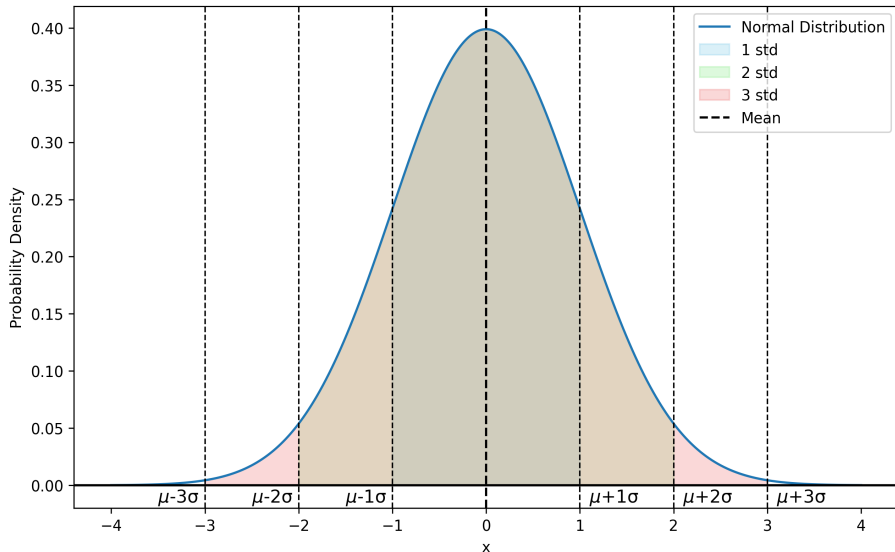
Properties of the Normal Distribution

Some key properties of the Normal distribution:

- **Symmetry:** The distribution is symmetric around the mean (μ).
- **68-95-99.7 Rule:**
 - About 68% of the data falls within one standard deviation of the mean.
 - About 95% falls within two standard deviations.
 - About 99.7% falls within three standard deviations.
- **Asymptotic:** The tails of the Normal distribution approach, but never touch, the horizontal axis.
- **Area under the Curve:** The total area under the Normal curve is 1.

Normal dist and standard deviation

Normal Distribution with Standard Deviations



Applications of Normal Distribution

The Normal distribution is used in a wide range of fields and applications:

- **Psychology and Education:** Test scores (e.g., IQ scores, SAT scores).
- **Finance:** Stock returns, asset prices.
- **Healthcare:** Measurements of physical traits like height, weight, and blood pressure.
- **Manufacturing:** Quality control, measurement of defects.
- **Natural Sciences:** Errors in scientific measurements, physical phenomena.

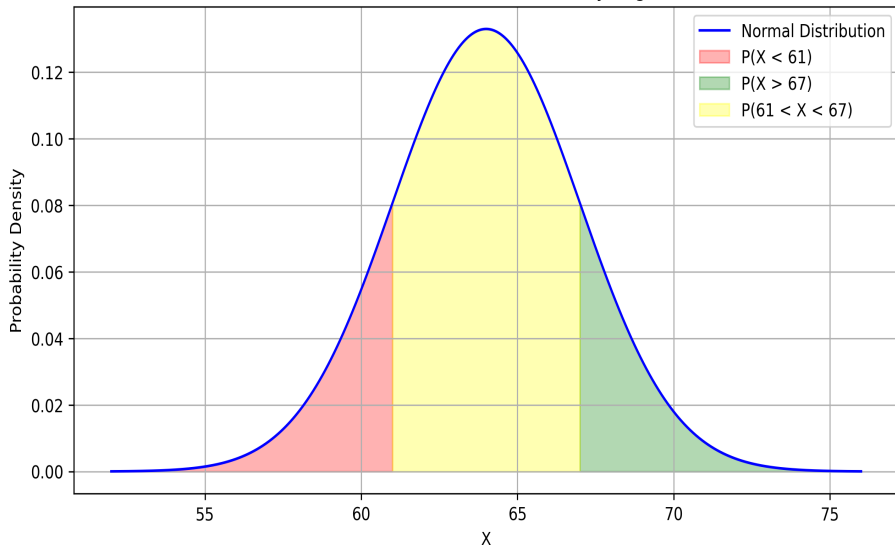
The normal distribution is often used in hypothesis testing, confidence intervals, and regression analysis.

Summary

- The Normal distribution is a continuous, symmetric distribution defined by its mean μ and standard deviation σ .
- It has a bell-shaped curve and is used to model many natural and social phenomena.
- The 68-95-99.7 Rule helps in understanding the spread of data in a Normal distribution.
- The Standard Normal distribution has $\mu = 0$ and $\sigma = 1$, and Z-scores are used to standardize data.
- It is widely applied in statistics, finance, healthcare, and more.

Normal Distribution

Normal Distribution with Probability Regions



Probability Distributions: Standard Normal Distributions

What is the Standard Normal Distribution?

- A special case of the Normal distribution.
- It is a Normal distribution with a mean of 0 and a standard deviation of 1.
- Mathematically, the Standard Normal distribution is denoted by:

$$Z \sim N(0, 1)$$

where:

- Z is the random variable.
- The mean (μ) is 0.
- The standard deviation (σ) is 1.

The probability density function (PDF) of the Standard Normal distribution is:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

where z is the standard score (also called the Z-score).

Standard Normal Distribution

Z-Score:

$$z = \frac{x - \mu}{\sigma}$$

Applications:

- **Standardization:**

- Many normally distributed datasets do not follow a Standard Normal distribution.
- Transform them into a Standard Normal form (using Z-scores).

- **Z-Scores:**

- Gives how many standard deviations a data point is from the mean.
- Used for identifying outliers and comparing data points from different distributions.

- **Simplicity in Calculation:**

- Well tabulated, simplifies calculations in hypothesis testing, confidence intervals, and many other statistical methods.

- **Central Limit Theorem (CLT)**

Z-Scores and Their Importance

The Z-score is a key component of the Standard Normal distribution. It measures the relative position of a data point within a distribution. The Z-score is computed as:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x is the data point.
- μ is the mean of the distribution.
- σ is the standard deviation of the distribution.

Interpretation of Z-scores:

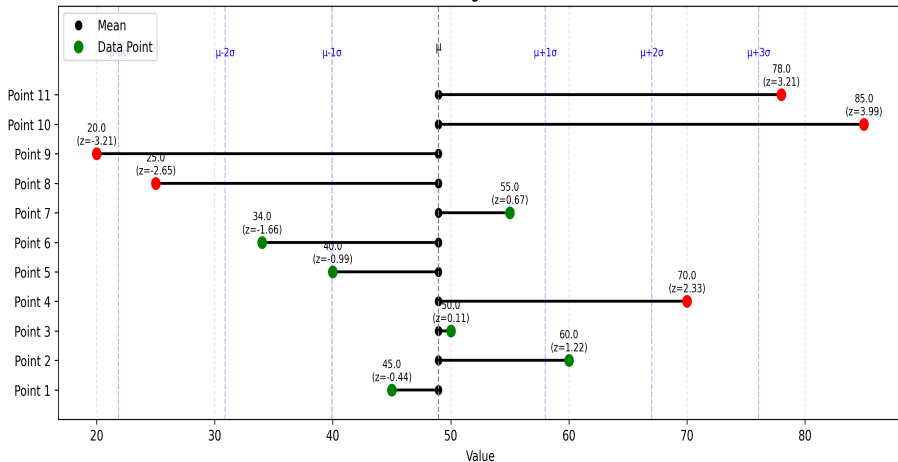
- A Z-score of 0 means that the data point is exactly at the mean.
- A positive Z-score indicates the data point is above the mean.
- A negative Z-score indicates the data point is below the mean.

Z-scores are essential for:

- Comparing scores from different normal distributions.
- Identifying outliers.

Normal Distribution

Z-Scores: Visualizing Distance from Mean

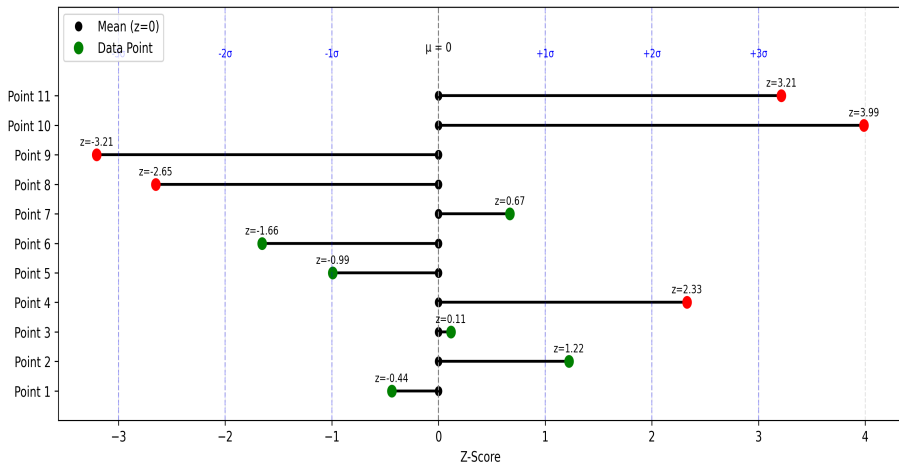


Observations from Z-Score Visualization

- **Mean (μ)** is shown as a black dot on each line and marked with a vertical dashed line for reference.
- Each data point is plotted on its own horizontal line, showing its distance from the mean.
- **Z-scores** are annotated beside each point, indicating how many standard deviations away the point is from the mean.
- Points within $\mu \pm 2\sigma$ are marked in **green**, indicating they fall within the typical range of a normal distribution.
- Points beyond $\mu \pm 2\sigma$ are marked in **red**, indicating they are **statistical outliers**.
- Vertical dashed lines mark standard deviation levels: $\mu \pm 1\sigma$, $\mu \pm 2\sigma$, and $\mu \pm 3\sigma$, helping to contextualize where each point lies.
- A point near 85 lies just beyond $\mu + 3\sigma$, a rare event (occurs in less than 0.15% of data in a standard normal distribution).
- Similarly, points like 20 and 25 fall below $\mu - 2\sigma$, also considered rare in a normal distribution.

Standard Normal Distribution

Z-Scores on Standard Normal Scale



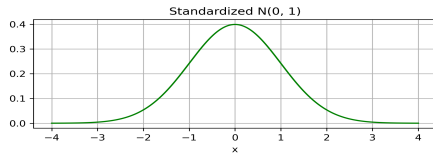
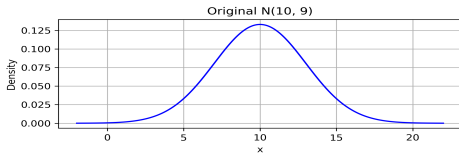
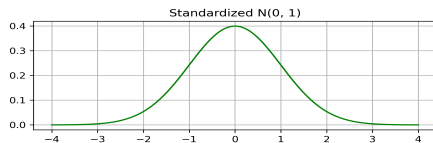
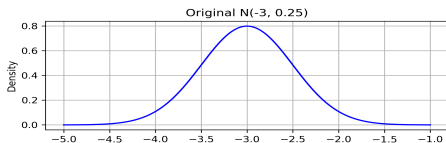
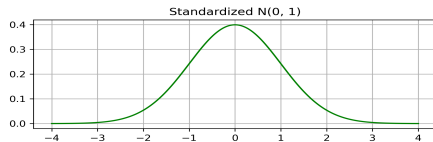
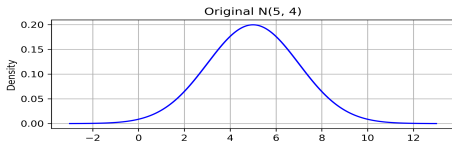
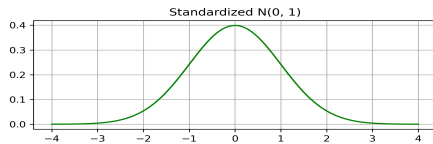
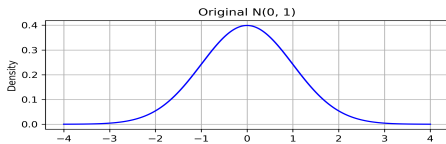
Observations from Standard Normal Z-Score Plot

- The mean (μ) of the standard normal distribution is at $z = 0$, marked with a vertical line.
- Each data point's position is shown as a z-score, indicating how far it lies from the mean in units of standard deviation.
- **Green points** represent values within $\pm 2\sigma$, considered typical or common.
- **Red points** indicate data that lies beyond $\pm 2\sigma$, considered outliers in a standard normal distribution.
- Vertical dashed lines mark $z = \pm 1, \pm 2, \pm 3$, helping students understand how data is spread around the mean.
- The standard normal scale provides a universal reference to compare different datasets, regardless of their original units.
- The transformation helps visualize the **relative rarity** of extreme points.
- For example, the point at $z > +3$ is extremely rare, appearing in less than 0.15% of standard normal data.

Comparison: Normal vs Standard Normal Representation

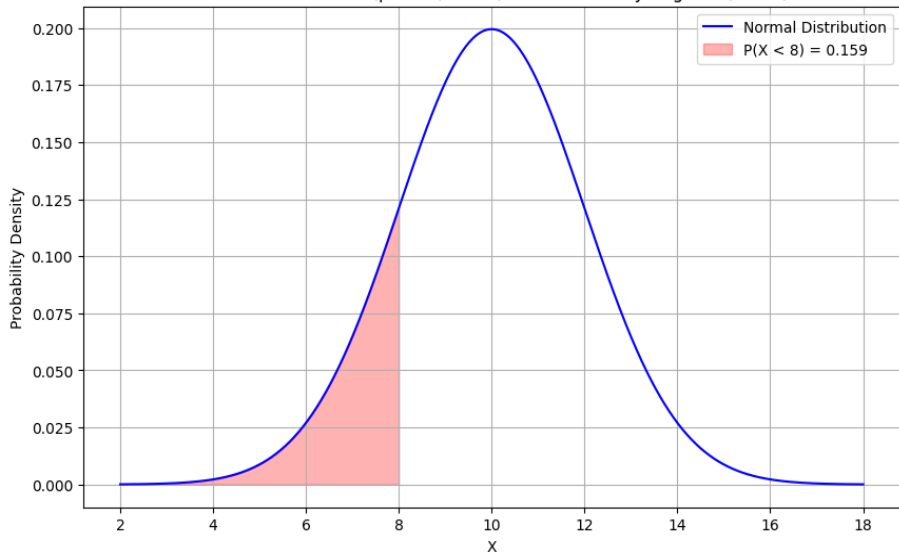
Feature	Normal Distribution Plot	Standard Normal Plot
X-Axis Scale	Original values (e.g., 45, 60, 85)	Standardized z-scores (z)
Mean Location	Varies with data (e.g., $\mu \approx 50$)	Fixed at $z = 0$
Interpretation of Distance	Contextual: depends on unit scale	Universal: measured in σ
Outliers Detection	Based on absolute value range	Based on $ z > 2$ or $ z > 3$
Standard Deviation Lines	$\mu \pm 1\sigma$, $\mu \pm 2\sigma$, etc.	$z = \pm 1$, ± 2 , ± 3
Visualization Purpose	Shows actual data spread and units	Shows relative spread across datasets
Color Coding	Based on deviation from mean	Same (based on z magnitude)

Standardization



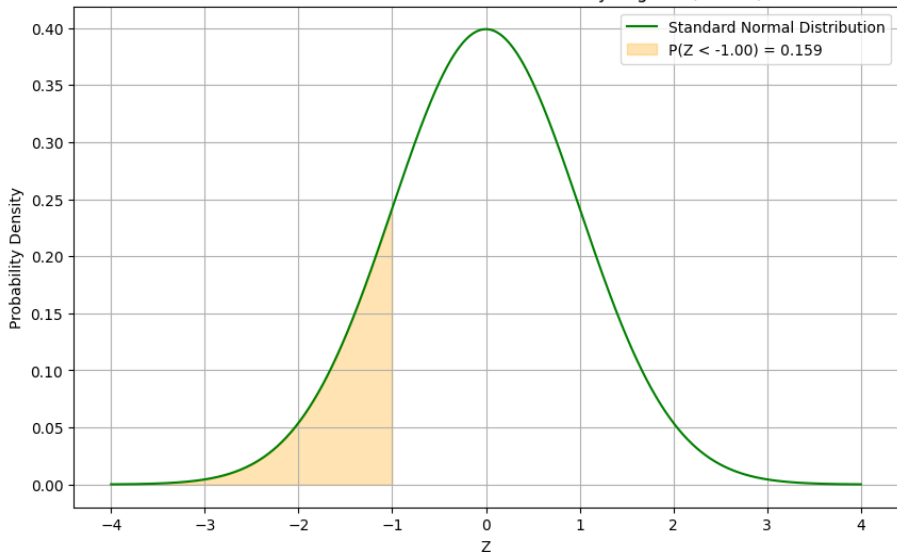
Normal distribution curve

Normal Distribution ($\mu = 10, \sigma = 2$) with Probability Region $P(X < 8)$



Standard Normal distribution curve

Standard Normal Distribution with Probability Region $P(Z < z_8)$



Standard Normal Table (Z-Table)

- The Standard Normal CDF is tabulated for various Z-scores.
- These tabulated values give the probability that a random variable is less than or equal to a given value in the Standard Normal distribution.
- For example, the Z-score table gives us:

$$P(Z < 1.96) = 0.9750$$

This means that the probability of a random variable being less than 1.96 in the Standard Normal distribution is 97.5%.

Z-tables are essential for:

- Finding probabilities for areas under the Normal curve.
- Hypothesis testing and confidence interval calculations.
- Determining critical values for statistical tests.

Application of the Standard Normal Distribution

The Standard Normal distribution is used in various areas of statistics and data analysis:

- **Hypothesis Testing:** In hypothesis testing, we standardize the test statistic (such as the Z-test) to compare it against the Standard Normal distribution.
- **Confidence Intervals:** The Z-distribution is used to calculate confidence intervals for population parameters (e.g., mean) when the population standard deviation is known.
- **Central Limit Theorem (CLT):** The Standard Normal distribution is a reference distribution in CLT for approximating the sampling distribution of the sample mean.
- **Data Transformation:** Converting a dataset into a Z-score (Standardizing) allows comparison across different datasets or variables.
- **Error Detection:** Z-scores are useful for identifying outliers in data, i.e., data points that are significantly higher or lower than the mean.

Finding Probability Between Two Heights Using the Standard Normal Distribution

- Heights of adult women follow a normal distribution with mean $\mu = 64$ inches and standard deviation $\sigma = 3$ inches.
- We want to find the probability that height lies between 61 and 67 inches.
- First, convert to Z-scores: $z_1 = \frac{61-64}{3} = -1$, $z_2 = \frac{67-64}{3} = 1$
- Use the Standard Normal Table or calculator to find CDF values.
- $P(Z \leq -1) = 0.1587$, $P(Z \leq 1) = 0.8413$
- Subtract: $P(61 \leq X \leq 67) = P(z_2) - P(z_1) = 0.8413 - 0.1587$
- Final Answer: $P(61 \leq X \leq 67) = 0.6826$, or about ****68.26%****

Why Compare Scores Using Standard Normal?

- Raw scores come from different normal distributions.
- These distributions may have different:
 - Means (μ)
 - Standard deviations (σ)
- A raw score of 75 might mean different things in different contexts.
- **Solution:** Convert to a common scale using the **standard normal distribution**.

Z-Score Formula:

$$z = \frac{X - \mu}{\sigma}$$

- X : Raw score
- μ : Mean of the distribution
- σ : Standard deviation
- z : Standardized score (on the standard normal distribution)

Interpreting Z-Scores

- $z = 0$: Score is exactly average
- $z > 0$: Score is above average
- $z < 0$: Score is below average
- The larger $|z|$, the further from the mean

Example: Comparing Two Students

Student A

- Score = 85, Mean = 70, SD = 10
- $z = \frac{85-70}{10} = 1.5$

Student B

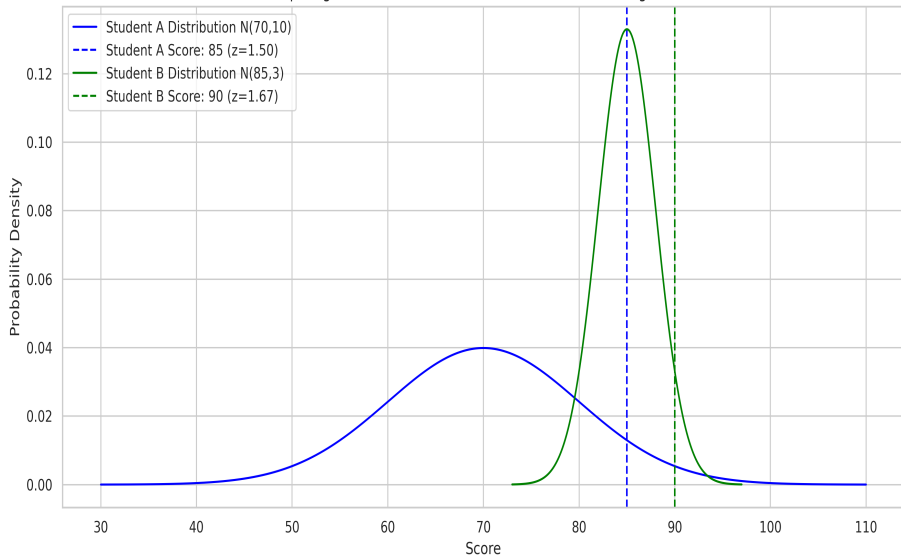
- Score = 90, Mean = 85, SD = 3
- $z = \frac{90-85}{3} = 1.67$

Conclusion

- Even though Student A had a higher raw score,
- Student B performed **relatively better** in their distribution, **More far away from average**.
- **Z-scores** make fair comparisons possible across different normal distributions.

Comparing scores from different normal distributions

Comparing Scores from Different Normal Distributions using Z-Scores



Summary: Normal Distribution Functions in Excel

Function	Purpose	Typical Use Case
NORM.DIST(x, mean, sd, TRUE)	CDF	Cumulative probability $P(X \leq x)$
NORM.DIST(x, mean, sd, FALSE)	PDF	Height of normal curve at x
NORM.S.DIST(z, TRUE)	Standard Normal CDF	$P(Z \leq z)$ from z-table
NORM.S.DIST(z, FALSE)	Standard Normal PDF	Height at z on standard normal curve
NORM.INV(prob, mean, sd)	Inverse CDF	Returns x such that $P(X \leq x) = \text{prob}$
NORM.S.INV(prob)	Inverse Std Normal	Returns z such that $P(Z \leq z) = \text{prob}$
STANDARDIZE(x, mean, sd)	Z-Score	Converts raw score x to $z = \frac{x-\mu}{\sigma}$

Tip: Use STANDARDIZE to normalize values and apply standard normal functions easily.

Probability Distributions: Chi-Square Distribution

Chi-Square Distribution: Overview

- The **Chi-Square distribution** is a continuous distribution used in hypothesis testing.
- Arises as the sum of squares of k independent standard normal variables:

$$X = Z_1^2 + Z_2^2 + \cdots + Z_k^2$$

- It is parameterized by the **degrees of freedom** k .
- **Mean** = k , **Variance** = $2k$
- Skewed right for small k ; becomes more symmetric as k increases.

PDF of Chi-Square distribution with k degrees of freedom:

$$f(x; k) = \frac{x^{(k/2)-1} e^{-x/2}}{2^{k/2} \Gamma(k/2)}, \quad x \geq 0$$

Derivation Summary:

- Let $Z_1, Z_2, \dots, Z_k \sim N(0, 1)$ (independent).
- Then $X = \sum Z_i^2 \sim \chi_k^2$
- Example for $k = 2$:

$$X = Z_1^2 + Z_2^2 \sim \chi_2^2$$

Real-World Scenario: Factory Quality Control

- A factory monitors rod lengths daily.
- Each day's deviation is standardized into a z-score.
- From 4 days: $z_1 = 1.2, z_2 = -0.8, z_3 = 0.5, z_4 = -1.5$
- All $z_i \sim N(0, 1)$, independent

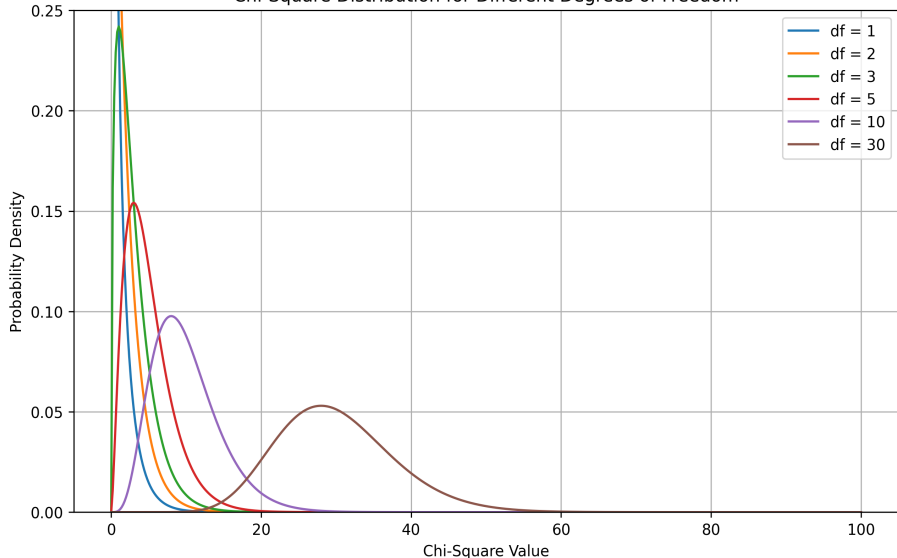
Constructing Chi-Square Statistic:

$$\chi_4^2 = z_1^2 + z_2^2 + z_3^2 + z_4^2 = 4.58$$

Applications and Interpretation

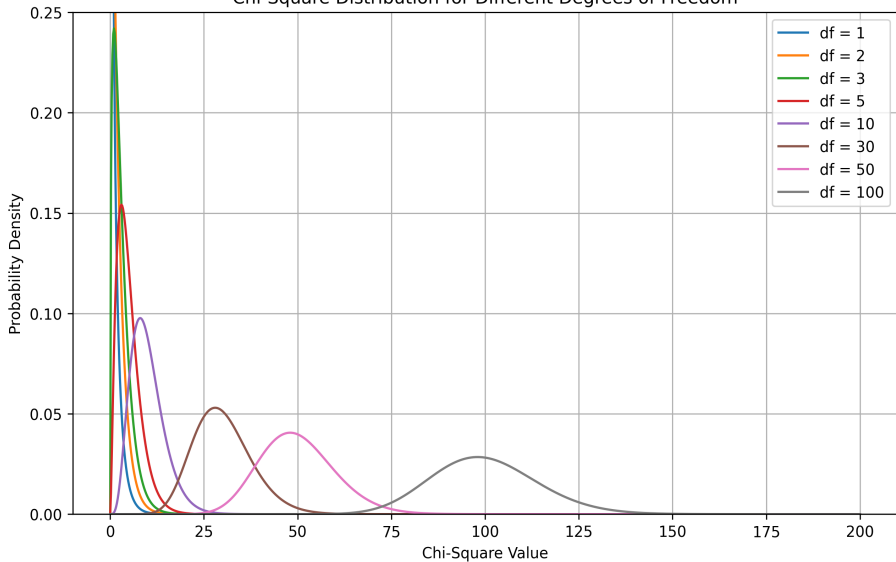
- Chi-Square tests are widely used for:
 - **Goodness-of-fit** tests
 - **Test of independence** in contingency tables
 - **Tests for variance** of a population
- A large χ^2 value may indicate:
 - Data does not follow expected distribution (reject H_0)
 - Or unusually high variability
- The critical value depends on the degrees of freedom and significance level.

Chi-Square Distribution for Different Degrees of Freedom



- For very large degrees of freedom, the distribution approximates a normal distribution.

Chi-Square Distribution for Different Degrees of Freedom



Probability Distributions: t-Distribution

What is the t-Distribution?

- The t-distribution is used instead of the normal distribution when:
 - The **population standard deviation** σ is **unknown**.
 - The **sample size** is **small**.
- It has heavier tails than the normal distribution.

When Do We Use the t-Distribution?

- When we estimate the population mean from a small sample.
- And when population standard deviation is unknown.
- We use the sample standard deviation, which adds uncertainty.
- t-distribution adjusts for this with degrees of freedom.

t-Statistic vs z-Statistic

Definition of the t-Statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- $\bar{x}$: sample mean
- μ : population mean (under null hypothesis)
- s : sample standard deviation
- n : sample size

Why Not Use the z-Statistic?

- z-statistic assumes population std. dev. σ is known:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

- In practice, σ is usually unknown.
- We estimate σ using sample std. dev. s .
- This adds uncertainty → use t-distribution instead.

Probability Density Function (PDF) of the t-Distribution

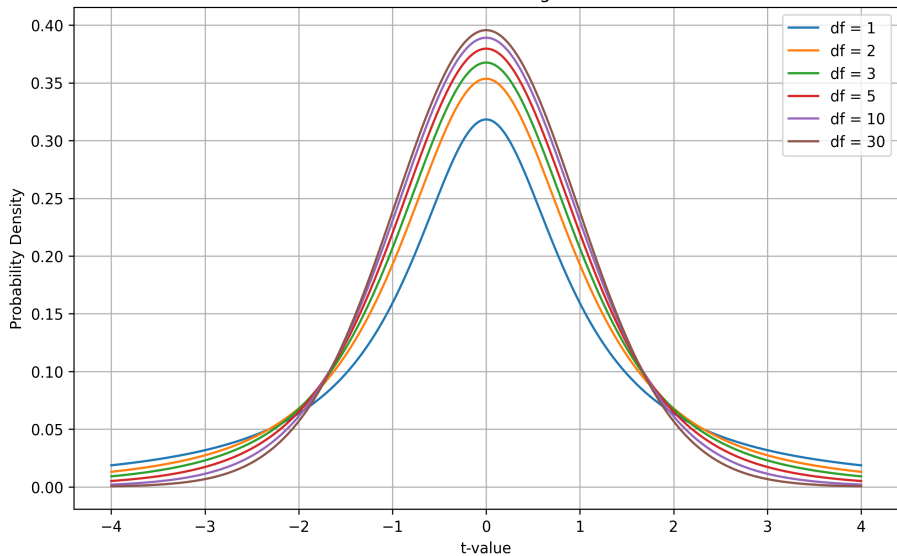
The probability density function (PDF) of the t-distribution is given by:

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Where:

- ν is the degrees of freedom,
- Γ is the Gamma function.

t-Distribution for Different Degrees of Freedom



Probability Distributions:F - Distribution

Introduction to F-Distributions

- The **F-distribution** is a **continuous probability distribution** that arises from the **ratio** of two independent chi-squared variables.
- It is used primarily in hypothesis testing, especially for comparing variances in two populations.
- The F-distribution is not symmetric, and it only takes positive values. So it is right-skewed
- Define the random variable F as the ratio of scaled chi-squared variables:

$$F = \frac{X_1/df_1}{X_2/df_2}$$

- As df_1 and df_2 increase, the distribution becomes more symmetric.
- It is defined by two parameters: degrees of freedom:

$$F_{m,n} \text{ where } m = df_{\text{numerator}}, n = df_{\text{denominator}}$$

Real-World Scenario: Comparing Two Machines

- A company uses two machines (A and B) to make ball bearings.
- Each machine is sampled for diameter variability.
- Measurements:
 - Machine A: 10 bearings $\rightarrow$ sample variance $s_1^2 = 0.015$
 - Machine B: 8 bearings $\rightarrow$ sample variance $s_2^2 = 0.008$

Constructing the F-Statistic

Formula

$$F = \frac{s_1^2}{s_2^2}$$

- Machine A (numerator): $s_1^2 = 0.015$
- Machine B (denominator): $s_2^2 = 0.008$
- Compute:

$$F = \frac{0.015}{0.008} = 1.875$$

Degrees of Freedom

- Sample sizes:

$$n_1 = 10, \quad n_2 = 8$$

- Degrees of freedom:

$$df_1 = n_1 - 1 = 9, \quad df_2 = n_2 - 1 = 7$$

- The F-statistic:

$$F = 1.875 \sim F(9, 7)$$

- This value is from an F-distribution with 9 and 7 degrees of freedom.

Understanding Alpha and Critical F-Value

- The symbol α (alpha) stands for a probability, like 0.05 or 5
- It shows how much area is in the right tail of the F-distribution curve.
- The critical F-value is a number that marks where this area begins.
- We write the critical value as:

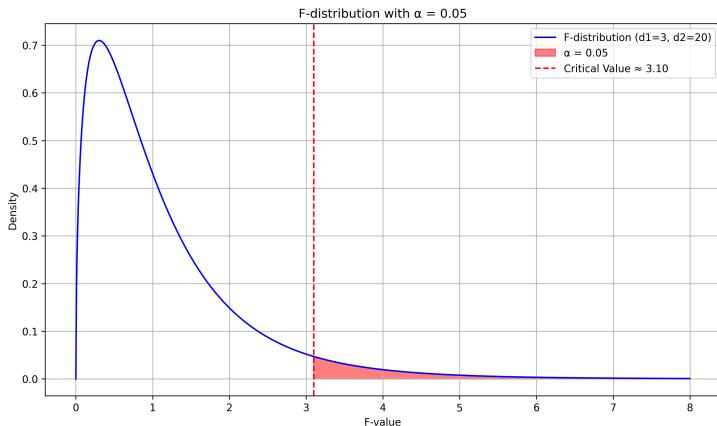
$$F_{\alpha,m,n}$$

- This means there's a probability of α (like 5%) that a value of F is greater than this number:

$$P(F > F_{\alpha,m,n}) = \alpha$$

F-Distributions

- The red area under the curve represents α .
- $F_{\alpha,m,n}$ is the value such that the area to its right equals α .
- Prob of $F_{3,20} < 3.10 = 1 - F_{0.05,3,20}$.



Reciprocal Identity of F-distributions

Fundamental Property:

$$F \sim F_{d_1, d_2} \quad \Rightarrow \quad \frac{1}{F} \sim F_{d_2, d_1}$$

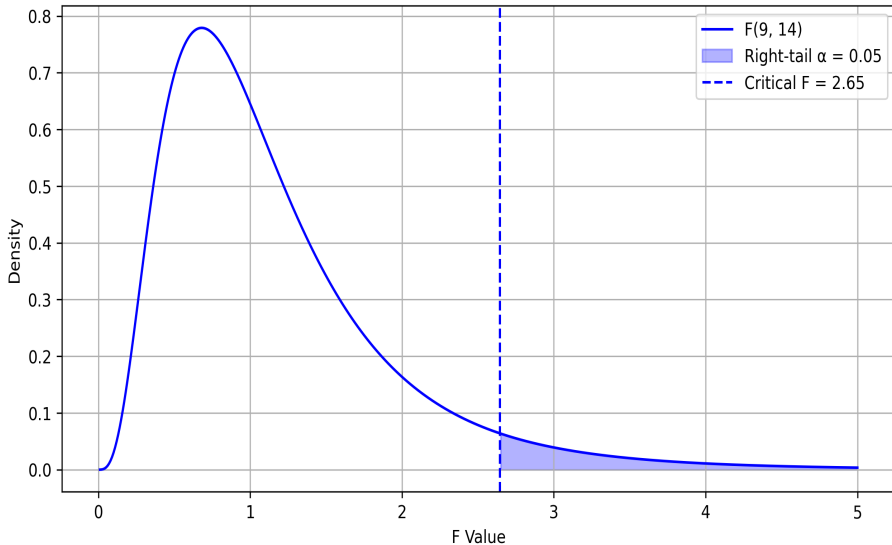
Implication on tail probabilities:

$$P(F_{d_1, d_2} > f) = P\left(F_{d_2, d_1} < \frac{1}{f}\right) = \alpha$$

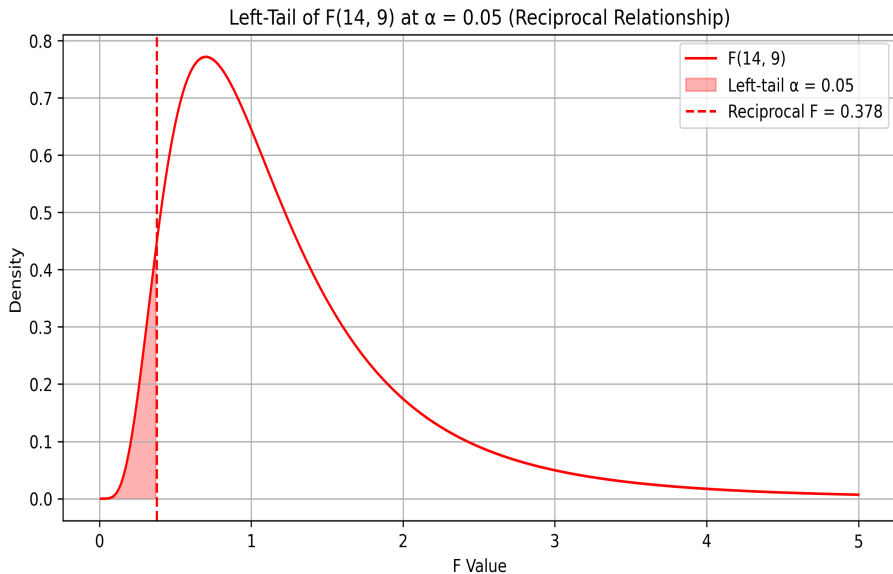
- A **right-tail** probability in one F-distribution becomes a **left-tail** probability in the reciprocal.
- Especially useful when test statistic is reversed.

F-distribution

Right-Tail of F(9, 14) at $\alpha = 0.05$



Reciprocal F-distribution



When to Use Each Distribution

- **Binomial:** Binary outcomes, fixed trials (e.g., coin flips, pass/fail tests).
- **Poisson:** Count of events in a fixed interval (e.g., arrivals per hour).
- **Normal:** Continuous data, central limit theorem applies (e.g., IQ scores).
- **Chi-Square:** Hypothesis testing for categorical data (e.g., independence test).
- **t-Distribution:** Small sample means, unknown variance (e.g., student test scores).
- **F-Distribution:** Comparing variances or ANOVA (e.g., multiple group comparisons).

Distributions Overview (I)

Distribution	Key Features	Parameters	Use Case / Shape
Binomial	Discrete, fixed trials	n, p	Count of successes, skewed or symmetric
Poisson	Discrete, rare events	λ	Rare events per interval, right-skewed
Normal	Continuous, bell curve	μ, σ	Natural data, Central Limit Theorem
Standard Normal	Special case of normal	–	z -scores, $\mu = 0, \sigma = 1$

Distributions Overview (II)

Distribution	Key Features	Parameters	Use Case / Shape
Chi-Square	Continuous, right-skewed	df	Variance tests, goodness-of-fit
t	Normal-like, heavier tails	df	Small samples, CI and tests on means
F	Ratio of variances	df_1, df_2	ANOVA, model comparison, right-skewed

Population and Sample

Population

- **Population** refers to the entire set of individuals, items, or data points that we are interested in studying.
- The population can be finite or infinite, and it contains all members from which we wish to draw conclusions.

Key metrics for a population include:

- **Total Population size** (N) ; Size of the population.
- **Population Mean** (μ): The average of all the values in the population.
- **Population Variance** (σ^2): Measures the spread or dispersion of the values in the population.
- **Population Standard Deviation** (σ): The square root of the population variance, representing the average distance of the data points from the population mean.

Sample

- A **sample** is a subset of the population that is selected for the purpose of statistical analysis.
- It is often impractical or impossible to collect data from an entire population.
- so samples are used to make inferences about the population.

Key metrics for a sample include:

- **Total Sample size** (n): Size of the sample.
- **Sample Mean** ($\bar{x}$): The average of all the values in the sample.
- **Sample Variance** (s^2): Measures the spread or dispersion of the values in the sample.
- **Sample Standard Deviation** (s): The square root of the sample variance, representing the average distance of the sample points from the sample mean.

Comparison: Population vs. Sample

The population and sample have similar metrics, but the key difference is that the population includes all members, while the sample is just a subset.

Metric	Population	Sample
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\hat{p}$

While the population parameters are fixed, the sample statistics are subject to sampling variability and serve as estimates of the population parameters.

Population Parameter vs. Sample Statistic

In statistics, we often distinguish between two key concepts:

- **Population Parameter**
- **Sample Statistic**

These concepts are fundamental to understanding how we make inferences about a population based on a sample of data.

Population Parameter

- A **population parameter** is a numerical value that summarizes a characteristic of an entire population.
- These parameters are fixed and unchanging.
- But they are often unknown because it's usually impractical to measure an entire population.

Examples of Population Parameters:

- **Population Mean** (μ)
- **Population Variance** (σ^2)
- **Population Standard Deviation** (σ)
- **Population Proportion** (p)

Key Point: Population parameters are fixed values, but they are often difficult or impossible to measure directly due to the size or accessibility of the population.

Sample Statistic

- A **sample statistic** is a numerical value calculated from the data of a sample, which is a subset of the population.
- Sample statistics are used to estimate population parameters

Examples of Sample Statistics:

- **Sample Mean** ($\bar{x}$)
- **Sample Variance** (s^2)
- **Sample Standard Deviation** (s)
- **Sample Proportion** ($\hat{p}$).

Key Point: Sample statistics are variable, meaning they can change from sample to sample, even when taken from the same population.

Key Differences: Population Parameter vs. Sample Statistic

Aspect	Population Parameter
Definition	A numerical value that summarizes a characteristic of the entire population.
Notation	μ (mean), σ^2 (variance), σ (standard deviation), p (proportion)
Fixed or Variable?	Fixed and constant (if the population is known).
Accessibility	Often unknown because it's difficult to measure the entire population.
Usage	Used to describe the entire population.

Aspect	Sample Statistic
Definition	A numerical value calculated from the data of a sample.
Notation	$\bar{x}$ (mean), s^2 (variance), s (standard deviation), $\hat{p}$ (proportion).
Fixed or Variable?	Variable; changes depending on the sample.
Accessibility	Known and easy to calculate because it's based on the sample data.
Usage	Used to estimate the corresponding population parameter.

Inferential Statistics

Inferential Statistics: Estimating Population Parameters

Since it's often impractical to measure an entire population, we rely on **inferential statistics** to make predictions or inferences about population parameters based on sample statistics.

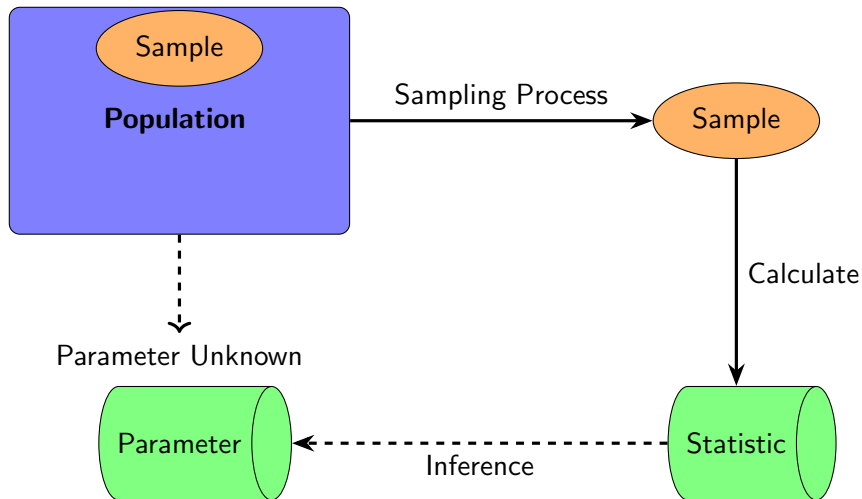
For example:

- If we want to know the average height of all high school students in a country, we might take a random sample of 500 students and calculate the sample mean height ($\bar{x}$).
- The sample mean ($\bar{x}$) can be used to estimate the population mean (μ).
- We can also compute confidence intervals and conduct hypothesis testing to make more precise inferences about the population parameter.

Key Point:

- Sample statistics are used as estimates of population parameters.
- There is always some degree of uncertainty associated with these estimates.
- This can be quantified using confidence intervals, margin of error etc.

Sampling and Inference



Sample Statistic: Its behaviour

- A **sample statistic** (e.g., sample mean $\bar{X}$, sample variance S^2) is computed from a sample drawn from a population.
- Each time we draw a sample, the values in the sample may change.
- Since the sample changes randomly, the statistic computed from it also changes.
- **Therefore, the statistic depends on random outcomes** — it is a function of random variables.
- **Conclusion:** A sample statistic is a **random variable** because its value varies across different samples.
- It means Sample statistics will have a distribution.

What is the average number of cups of coffee per day consumed by college students in the U.S.?

- Population: All college students in the U.S.
- Parameter of interest: Average daily coffee consumption

Why We Sample

- It's impractical to survey the entire population.
- We take a random sample to estimate the population mean.

Example:

Randomly select 30 students and ask how many cups of coffee they drink per day.

Sample Data (Example) [2, 0, 1, 3, 2, 1, 2, 4, 0, 1, 2, 3, 1, 2, 2, 1, 5, 1, 0, 3, 2, 1, 2, 0, 4, 3, 2, 1, 2, 1]

- Sample size: $n = 30$
- Sample mean: (computed visually)

Each Response is a Random Variable

- Let X_i = cups of coffee consumed by the i -th student.
- Each X_i is a random variable:
 - Depends on which student is randomly chosen out of total students.
 - Value varies with each sample.
- The sample mean $\bar{X} = \frac{X_1 + X_2 + \dots + X_{30}}{30}$ is also a random variable.

Summary Table:

Concept	Example
Population	All U.S. college students
Random Variable X_i	Coffee cups consumed by student i
Sample	30 values like [2, 0, 1, ..., 1]
Random selection	Selecting students at random
Why It's Random	Different students give different values

Central Limit Theorem (CLT)

Introduction

- The Central Limit Theorem (CLT) is a fundamental theorem in probability theory.
- It states that, under certain conditions, the sum (or average) of a large number of independent and identically distributed (i.i.d.) random variables follows a normal distribution, regardless of the original distribution.

Formal Definition

Let $X_1, X_2, \dots, X_n$ be a sequence of i.i.d. random variables with mean μ and variance σ^2 . Define the sample mean as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then, as $n \rightarrow \infty$,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

This means the standardized sample mean converges in distribution to a standard normal distribution.

Why is CLT Important?

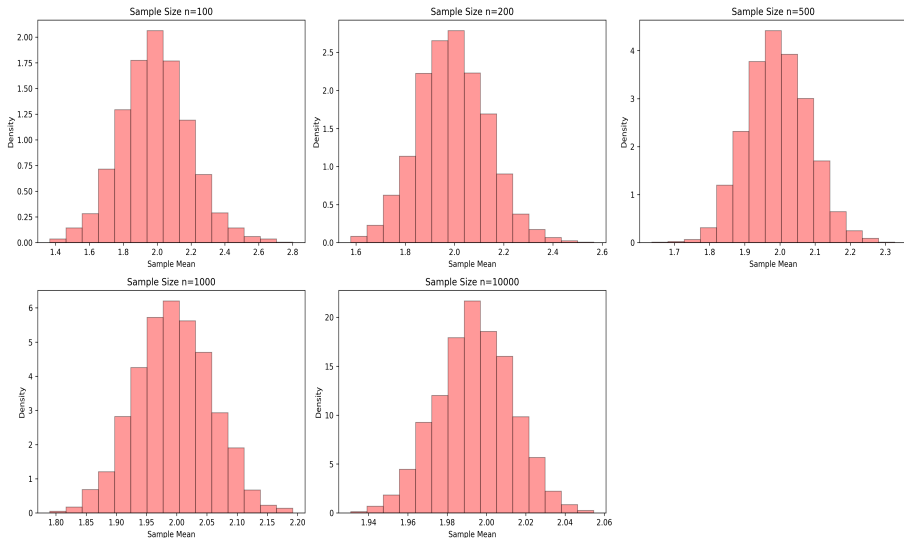
- Enables statistical inference when population distribution is unknown.
- Justifies the normality assumption in many practical applications.
- Forms the basis for confidence intervals and hypothesis testing.

Illustration: CLT in Action

- Consider different distributions (e.g., uniform, exponential, binomial).
- Take samples and compute sample means.
- Observe how the distribution of sample means approaches a normal distribution.
- Repeat for different sample sizes and observe the different distributions.

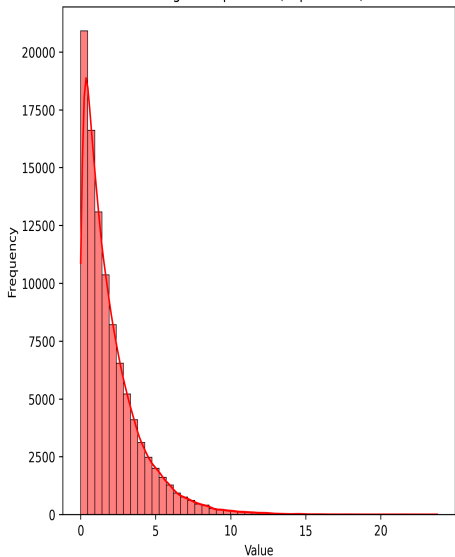
For 3000 sample means

Comparison of Sample Mean Distributions for Different Sample Sizes

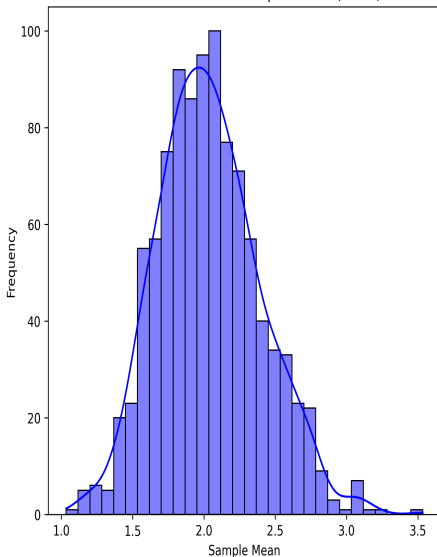


For sample size 30 and 1000 samples

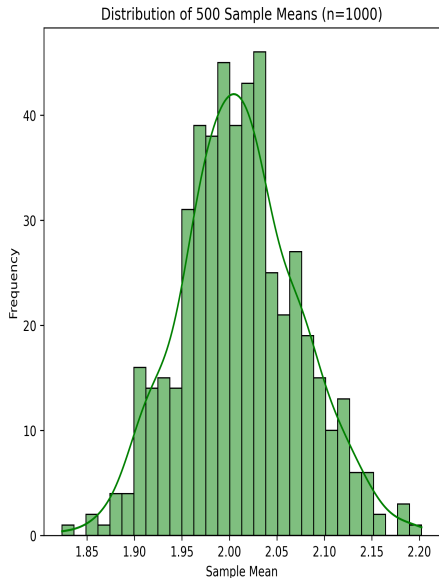
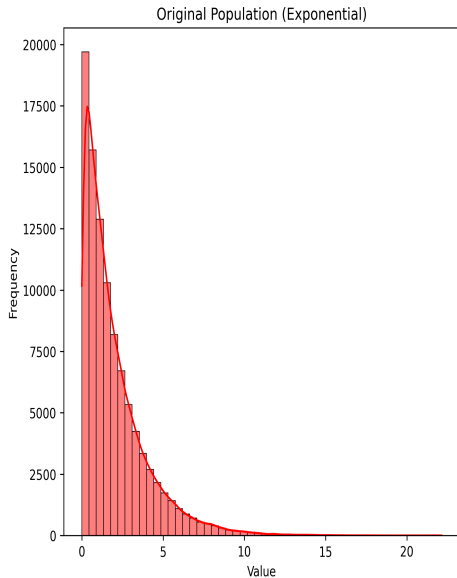
Original Population (Exponential)



Distribution of 1000 Sample Means ($n=30$)



For sample size 1000 and 500 samples



Effect of Sample Size vs. Number of Samples

- **Sample Size** n affects the **spread (variance)** of the sampling distribution.

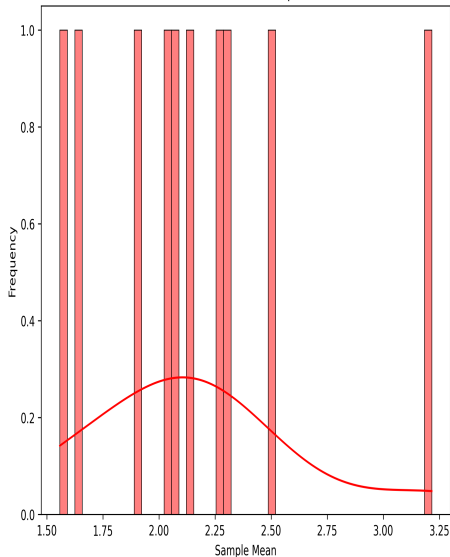
- According to the Central Limit Theorem:

$$\text{Standard Error} = \frac{\sigma}{\sqrt{n}}$$

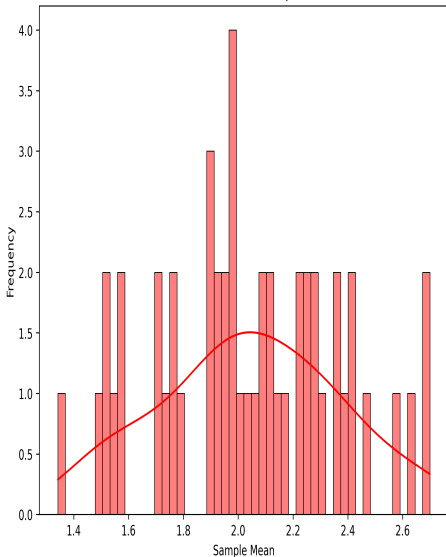
- As n increases, the standard error decreases.
 - The sampling distribution becomes narrower and more concentrated around μ .
- **Number of Samples** affects the **smoothness** of the sampling distribution curve.
 - More samples mean a smoother and more accurate estimate of the distribution of $\bar{X}$.
 - But it does **not** change the location or spread of the distribution itself.

For Fixed sample size

CLT Visualization - Samples: 10

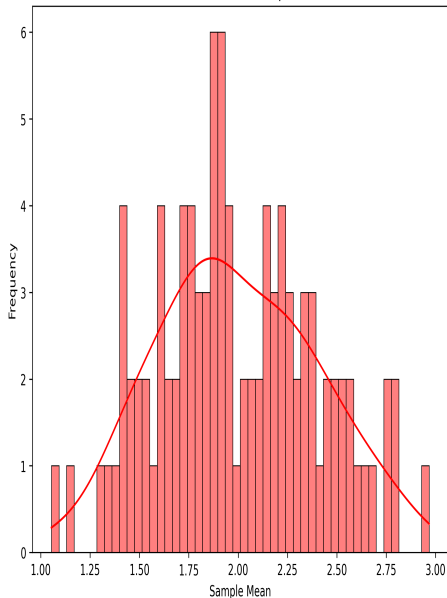


CLT Visualization - Samples: 50

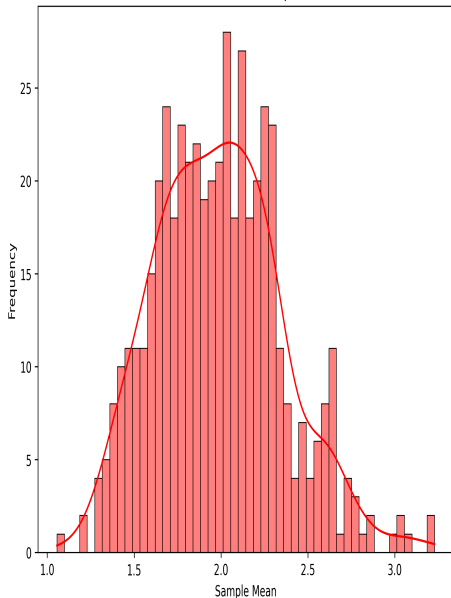


For Fixed sample size

CLT Visualization - Samples: 100

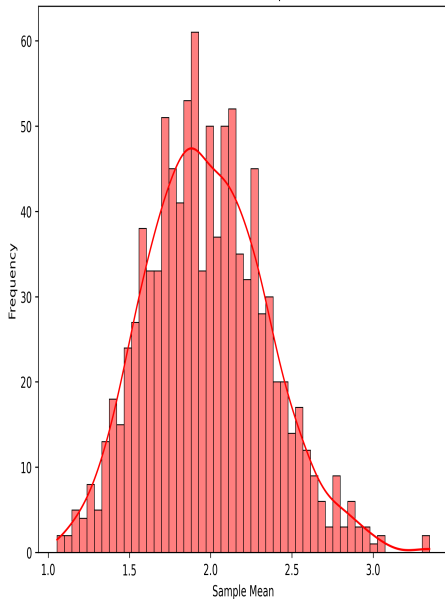


CLT Visualization - Samples: 500

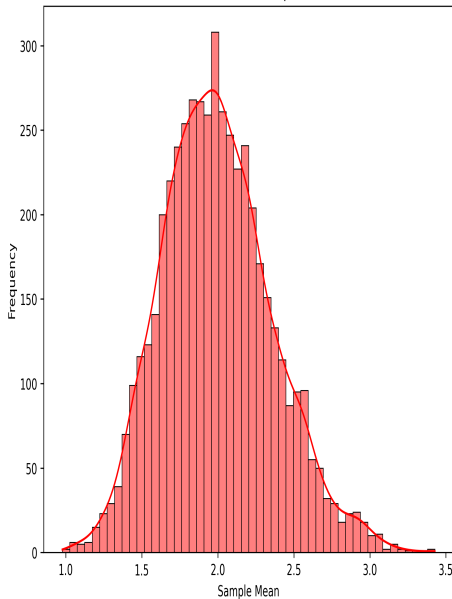


For Fixed sample size

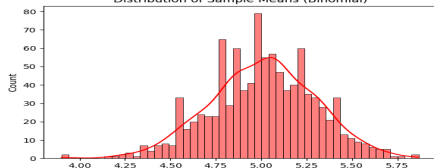
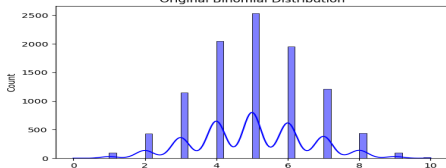
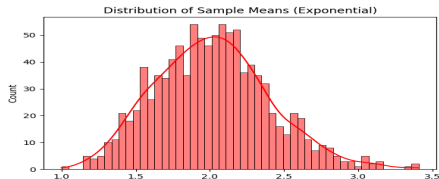
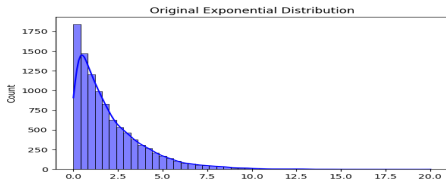
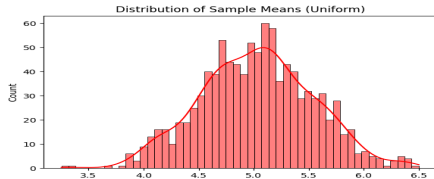
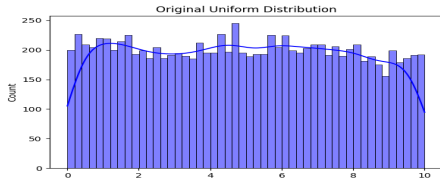
CLT Visualization - Samples: 1000



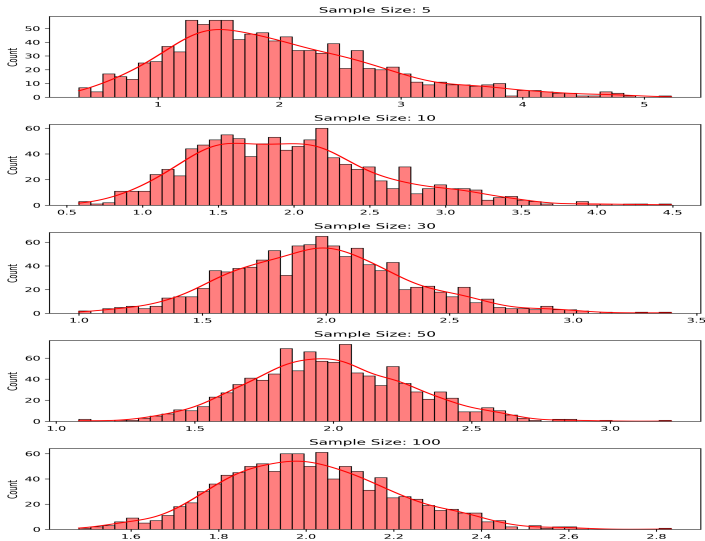
CLT Visualization - Samples: 5000



For any distribution



For different sample sizes



Effect of Sample Size and Number of Samples on CLT

- The **Central Limit Theorem (CLT)** states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases.
- **Sample size (n)** affects how closely the distribution of the sample mean resembles a normal distribution.
- The **number of samples** (in simulations) affects how smooth and stable the observed sampling distribution is.
- Larger sample sizes improve the normal approximation even for skewed or non-normal populations.

CLT: Practical Effects Summary Table

Factor	Effect on CLT	Explanation
Sample size (n)	Controls normality of sample mean distribution	Larger n gives better approximation to the normal distribution
Number of samples	Affects stability in simulations	More samples reduce simulation variability, but do not change CLT itself
Small n with normal population	Sample mean is still normal	Normal population ensures normal sample mean regardless of n
Small n with skewed population	Poor approximation	Sampling distribution is not normal; CLT doesn't hold well
Large $n \geq 30$	CLT applies generally	Works even for skewed or non-normal populations

Example Application

Suppose the heights of students in a university are not normally distributed. However:

- If we randomly select 30 students and compute the sample mean,
- The sample mean follows approximately a normal distribution.
- This allows us to make statistical inferences using normality-based methods.

Limitations and Assumptions

- CLT requires a sufficiently large sample size ($n \geq 30$ is a common rule of thumb).
- Assumes independence of random variables.
- Works best when individual observations have finite variance.

Does CLT Apply to Other Statistics?

Statistic	CLT Applies?	Notes
Sample mean $\bar{x}$	Yes – classic case	Most widely used
Sample proportion $\hat{p}$	Yes	Mean of 0s and 1s
Sample sum $\sum X_i$	Yes	Scales with sample size
Sample median, mode	Not directly	Needs different limit theorems
Sample variance	More complex	Converges to chi-square dist.

CONFIDENCE INTERVALS

What is a Confidence Interval?

- A **confidence interval (CI)** is a range of values estimated from sample data, likely to contain the true population parameter.
- Common confidence levels: **90%, 95%, 99%**.
- A higher confidence level → wider interval, more certainty.
- Example: “We are 95% confident the true mean lies between 50 and 60.”

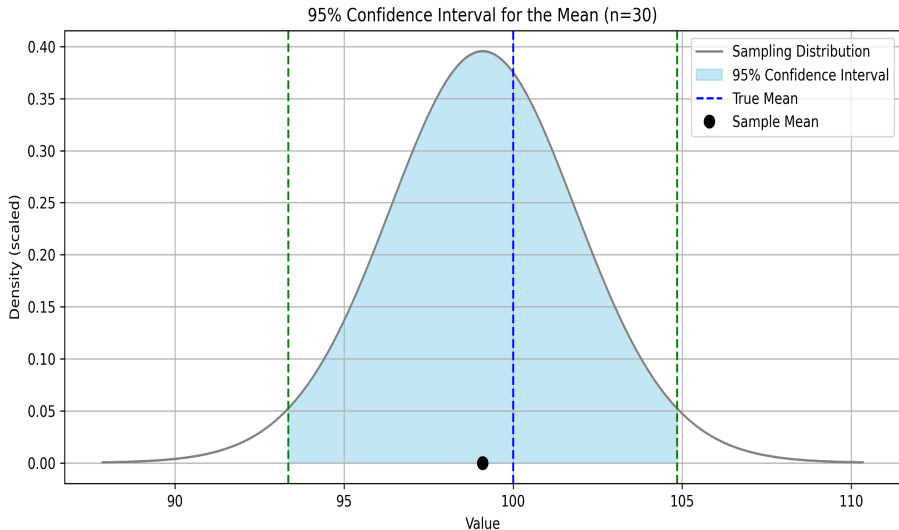
CLT and the Confidence Interval Formula

- The **Central Limit Theorem (CLT)**: For large n , the distribution of sample means $\bar{X}$ is approximately normal.
- Mean = μ , Standard Error = $\frac{\sigma}{\sqrt{n}}$
- This leads to the CI formula for known σ :

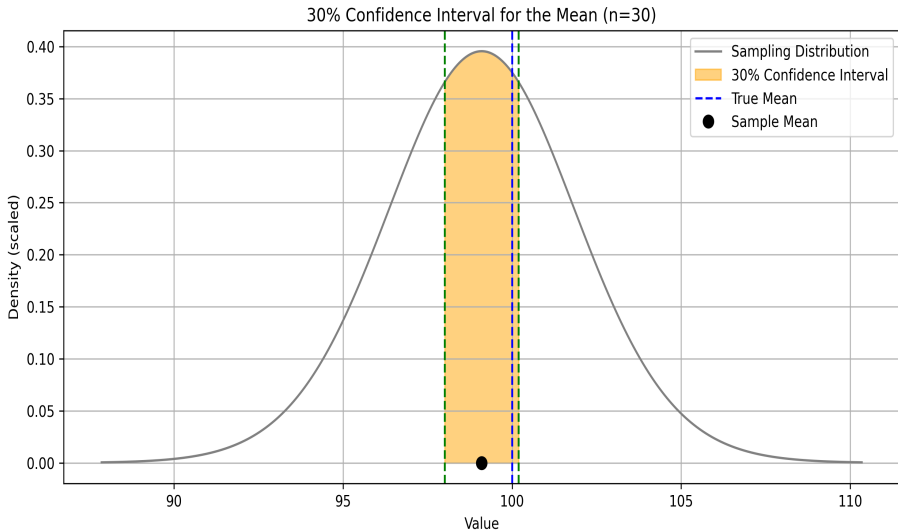
$$\bar{X} \pm Z \times \frac{\sigma}{\sqrt{n}}$$

- Common Z-values:
 - $Z = 1.96$ for 95% confidence
 - $Z = 2.576$ for 99% confidence

For One Sample

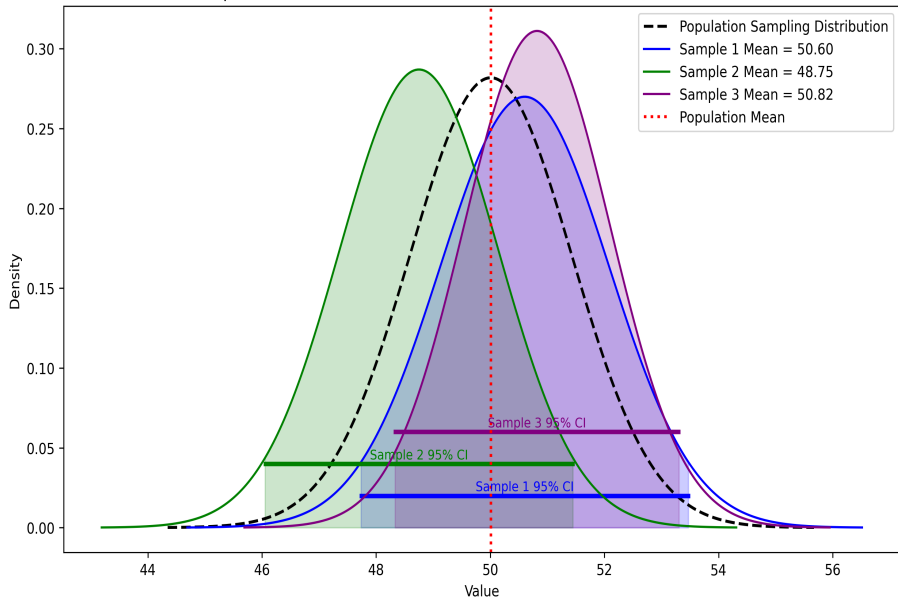


For One Sample



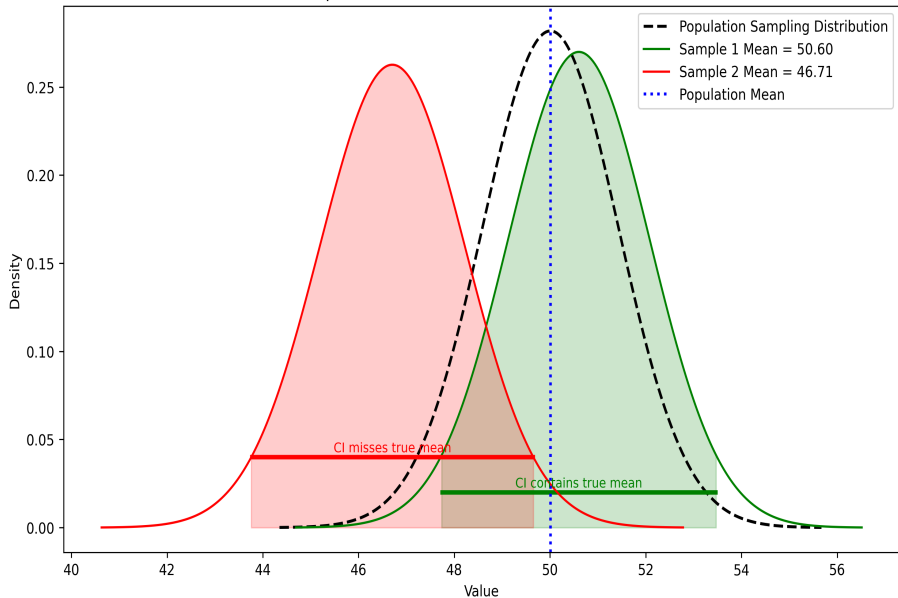
95% Confidence Interval

Sample Mean Distributions with 95% Confidence Intervals (Horizontal Lines)

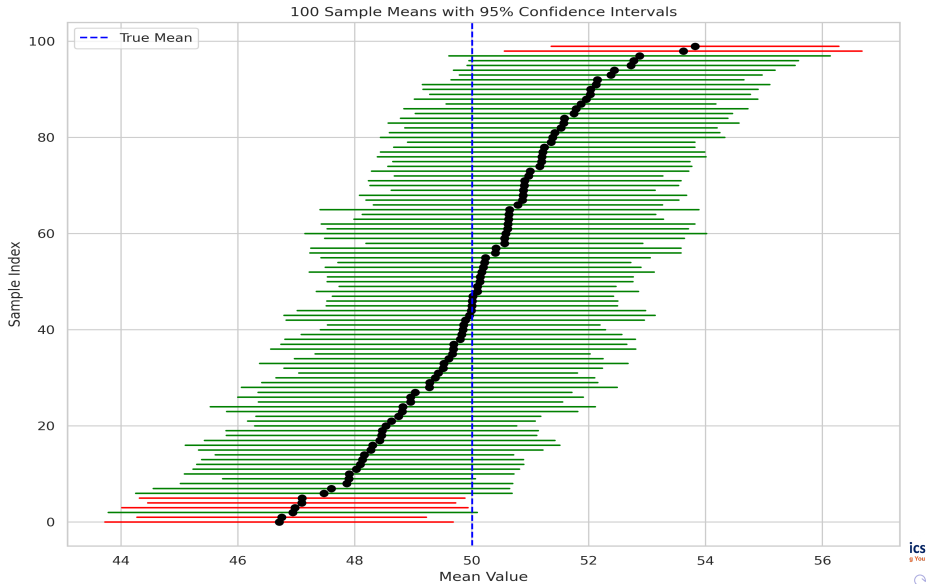


95% Confidence Interval

Two Sample CIs — One Contains True Mean, One Does Not

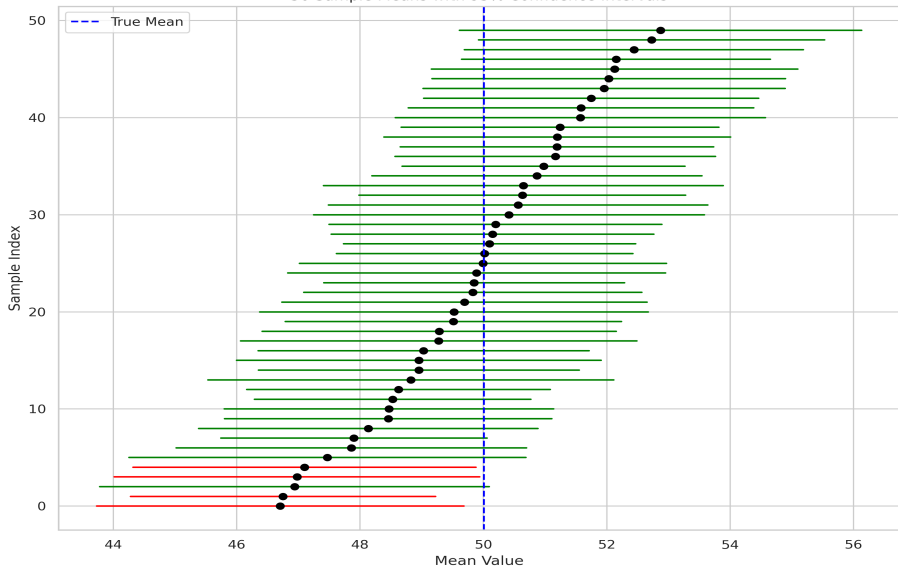


95% Confidence Interval



95% Confidence Interval

50 Sample Means with 95% Confidence Intervals



Interpretation of Confidence Interval

- A 95% CI means: If we took 100 samples, 95 of the CIs would contain the true mean.
- *It does not mean* there's a 95% chance that one specific CI contains the parameter.

What Affects the Width of a Confidence Interval?

- **Sample Size n :**

- Larger $n \rightarrow$ smaller $\frac{\sigma}{\sqrt{n}} \rightarrow$ narrower CI $\rightarrow$ more precision
- Smaller $n \rightarrow$ wider CI $\rightarrow$ more uncertainty

- **Confidence Level:**

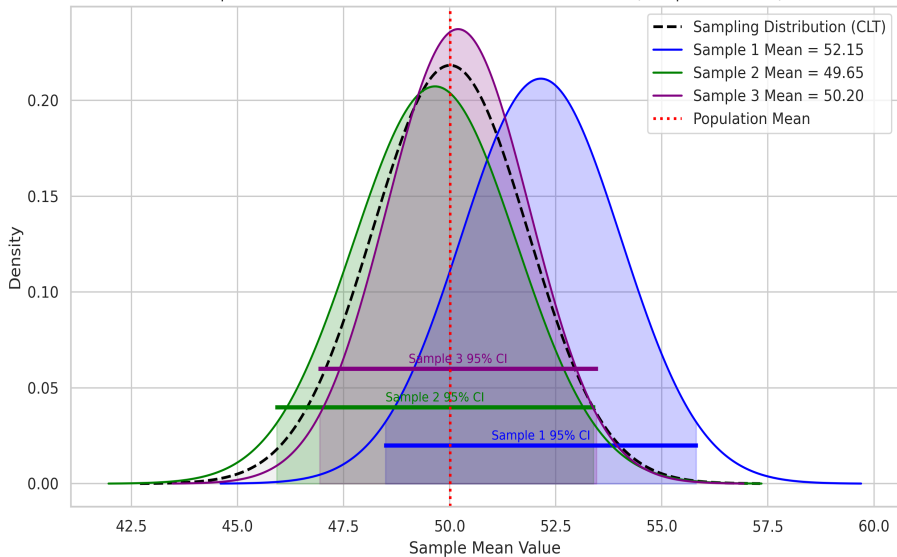
- Higher confidence (e.g., 99%) $\rightarrow$ wider interval
- Lower confidence (e.g., 90%) $\rightarrow$ narrower interval

- **Overlapping vs. Non-overlapping Intervals:**

- Overlapping $\rightarrow$ likely no significant difference
- Non-overlapping $\rightarrow$ suggests statistical significance

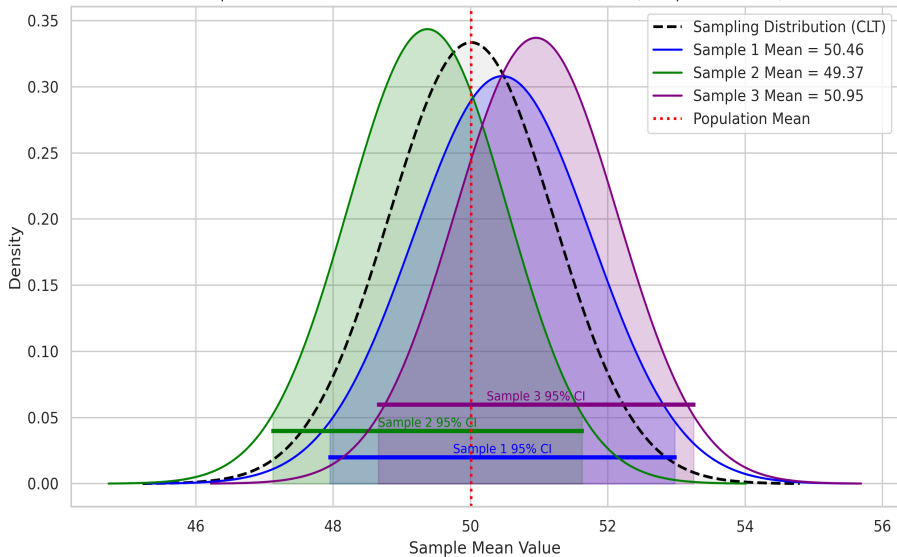
95% Confidence Interval

Sample Mean Distributions and 95% Confidence Intervals (Sample Size = 30)



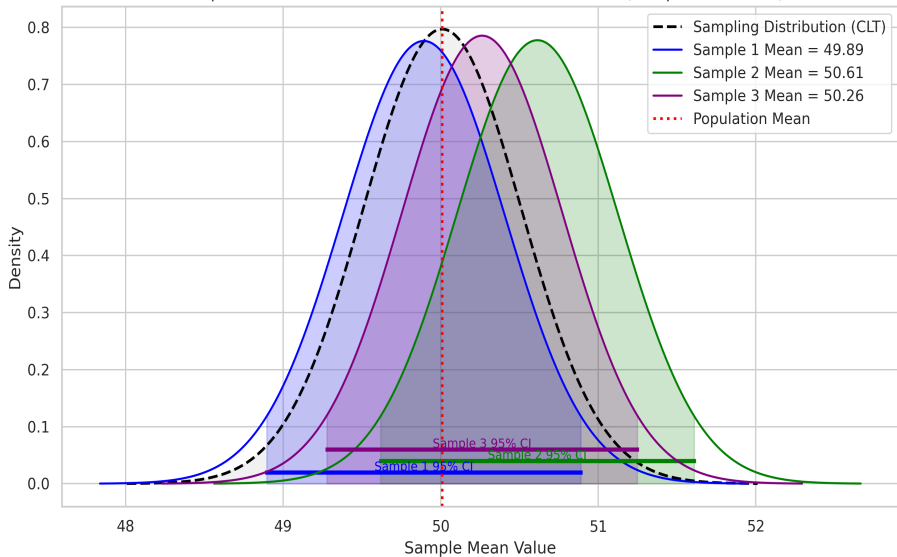
95% Confidence Interval

Sample Mean Distributions and 95% Confidence Intervals (Sample Size = 70)



95% Confidence Interval

Sample Mean Distributions and 95% Confidence Intervals (Sample Size = 400)



Confidence Intervals of Two Samples

- Confidence Intervals (CIs) for two samples help compare means or proportions between two groups.
- Useful in A/B testing, medical trials, education research, and more.
- Interpretation depends on how the two CIs behave:
 - **No Overlap:** Strong evidence of a significant difference between groups.
 - **Slight Overlap:** Possible difference, further testing (e.g., t-test) needed.
 - **Complete Overlap:** No significant difference inferred from data.
- Example: Comparing average scores of students in two teaching methods.

Real-World Examples: Confidence Intervals of Two Samples

- **No Overlap (Significant Difference)**

- *Medical Study:* Drug A vs Drug B
- **Conclusion:** Drug B significantly lowers blood pressure more than Drug A.

- **Partial Overlap (Uncertain Difference)**

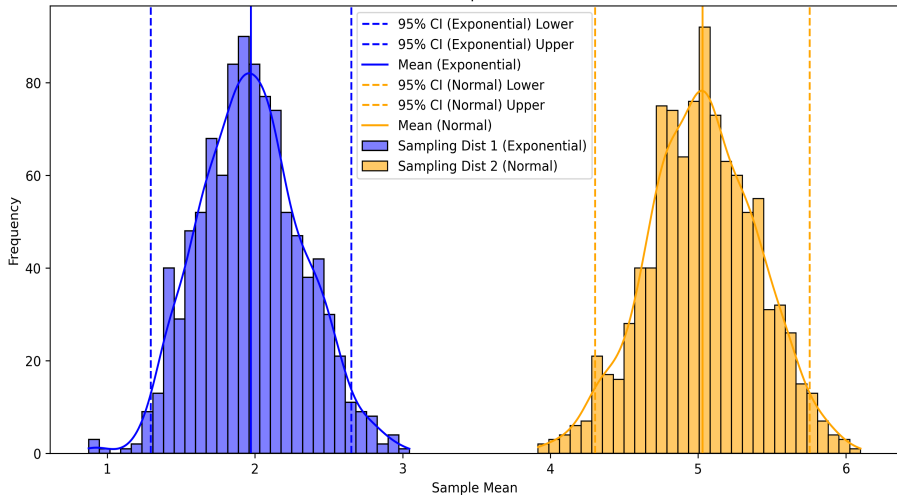
- *Education:* Online vs In-person teaching methods
- **Conclusion:** Possible difference in average scores; further testing (t-test) needed.

- **Complete Overlap (No Significant Difference)**

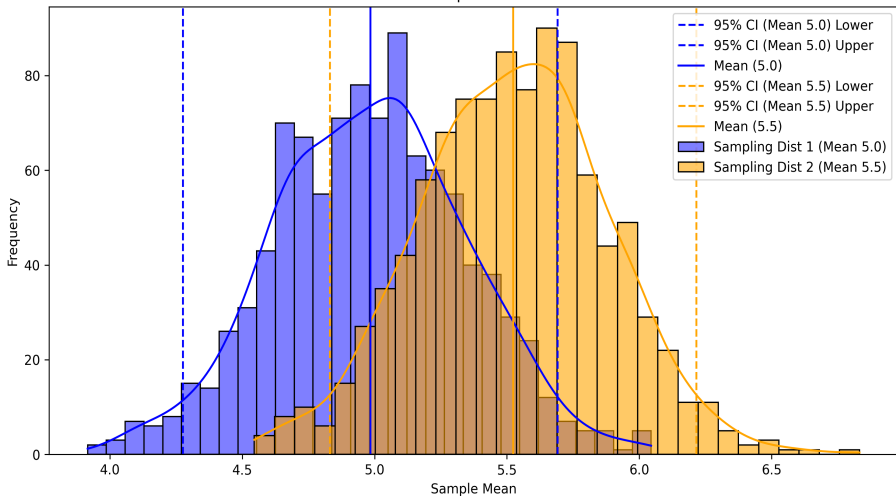
- *Marketing:* Two website layouts (A/B test)
- **Conclusion:** No clear evidence that one layout leads to more sales.

Confidence Intervals of Two Samples

Comparison of Sampling Distributions
No Overlap Detected



Comparison of Sampling Distributions Overlap Detected



Real-World Applications

- Confidence intervals are used in medical research to determine the effectiveness of treatments.
- In finance, they help estimate stock market trends.
- In quality control, they assess manufacturing consistency.

Limitations of Confidence Intervals

- Assumes random sampling; biased samples affect validity.
- Relies on the assumption that data follow a particular distribution.
- Wider intervals may not always provide useful conclusions.

Conclusion

- Confidence intervals provide a way to estimate population parameters with a degree of certainty.
- The width of the interval depends on sample size and confidence level.
- Proper interpretation is crucial for making informed statistical decisions.

THANK YOU